



TEMENOS

The software specialist for banking and finance

Objectives for the course

- Introducing the participants to the concept of payments hub
- Configuring and processing incoming, outgoing, redirect and book SWIFT payments
- Overview of batch processing (Single debit and multiple credits)
- Understand the need and use of warehousing capability, debit authority, code words and balance reservation
- Overview of linkage to STeP (Automated Repair Tool) and AML (Filtering)
- Understand clearing framework and how system can be scaled to support additional clearings
- Understand how the SEPA payments processing using ESB layer
- Understand how direct debits (both incoming and outgoing) are processed
- Using reports and enquiries to perform payments monitoring and analytics



Prerequisites for this Course

- T24 Induction course (T2ITC)
- Understanding of International and Domestic Payments Business
- Hands on experience with T24 product (or) Any other Banking / Payment Product

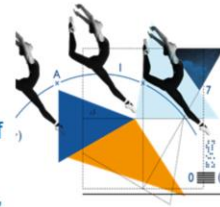
Preferable prerequisites for this course

TPH Day 1

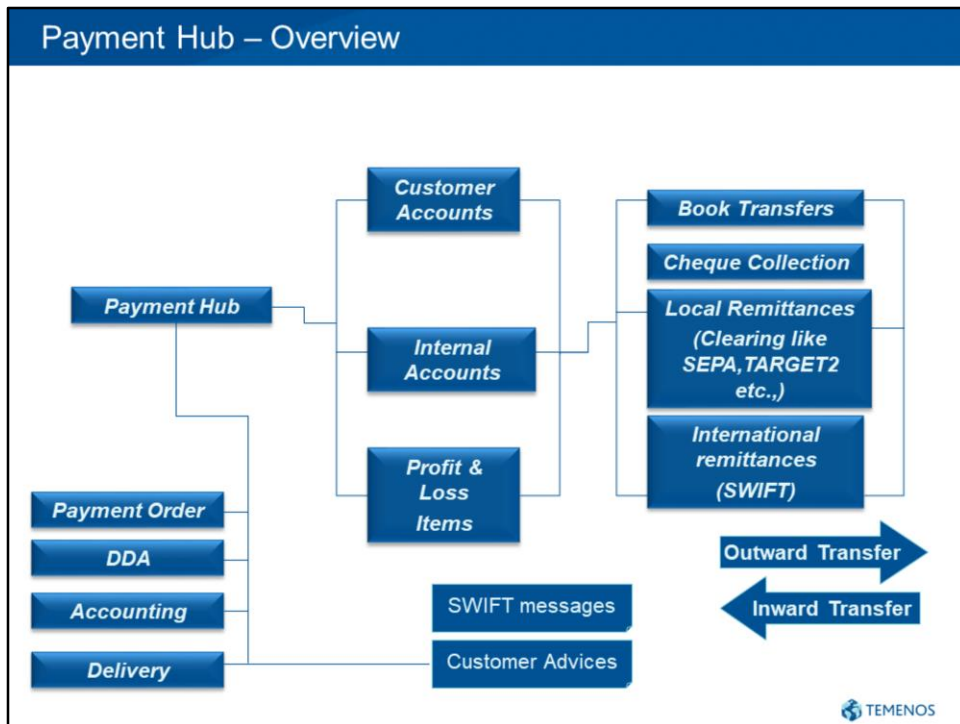
Learning Objectives

Objectives

- Learn about the:
 - Universal Payment Process Flow in Temenos Payment Suite
 - Understand the Data Model of TPH
- Learn How to setup the basic Static Data Configuration of TPH
- Understand how Incoming messages are received, accepted and mapped in TPH
- Understand the concept of
 - Weight Assignment
 - Debit Authority
 - Debit Party Determination
 - Warehousing of Payments
 - Balance Check



End of Day 1 – Learning objectives will be met.



Temenos Payments Hub (TPH) is a universal payments hub solution. It has a Universal Payments Process which enables centralization of all payment operations under a central infrastructure to support multiple payment formats and global payment mechanisms. The payment hub solution provides banks with a single solution to configure their payment processing for all customers, specifying different processing for high and low care customers by parameterization without requiring any software changes.

Internally, payments can be made to or from a Customer's account or an internal account. It could involve transferring money from one customer account to another customer account. Some of the external types of payment include Cheques, Mail Payment, Banker's Draft, Clearing House Payments, International payments via Correspondents etc.

Instructions can be fed manually or from other electronic communication systems like SWIFT. It is linked to all types of outputs whether paper or wire through the Delivery setup in T24.

If there is an interface with SWIFT, then SWIFT message can be generated for Outward Transfers. In respect of incoming SWIFT message it can be received and automated to process the payment.

The application is designed to handle all types of currencies, local or foreign and inward or outward payments.

On a real time basis, the application raises accounting entries and updates the core

limit setup whenever an account is debited or credited.

Payment Hub Dependencies

- Payment Hub makes use of
 - CUSTOMER
 - ACCOUNT
- Core dependencies
 - Delivery
 - Accounting
 - Currency & Holiday Table
- Payment Hub also uses other Static tables
- Payment Hub also work as a Standalone Product (communication with external system for Customer, DDA and GL)



We need Accounts to handle various customer type transactions say for movement of funds from debit accounts to credit accounts. We need to have Customer records before opening Customer type of accounts. Nostro accounts will be needed to transfer money externally using Agency arrangements.

Delivery in T24 refers to generation of Advices or Messages. Delivery is an integral part of Core in T24 which is closely linked to transactions input through Payment Hub module. Predefined messages/advices are generated on authorisation of transaction. The relevant messages are produced as per mapping and formatting setup such that details from the field input in transaction are mapped to the pre defined fields and SWIFT tags. They are sent through appropriate channels, Print or SWIFT. The channel or mode of delivery can be configured system-wide for each message type and also setup at the customer or account level.

On authorisation of transaction, accounting entries are generated.

When an account is debited, system checks for balance in the account. Limit, if any, sanctioned and attached to the account, is checked for availability of limit. Override messages are raised for overdraft above the limit availability.

Payment Hub application makes use of certain Static tables, like PP.CURRENCY, PP.COMPANY.PROPERTIES, PP.PROGRAMSPERWEIGHT, PP.REGION etc.

Benefits of Payment Hub

- TPH offers excellent breadth and depth of functionality as well as flexibility
- Payment Hub features offers ability to skip, prioritise and specify later date execution or override changes manually and manage service level agreements through parameterization
- TPH enables maximize Straight-through Processing (STP) with the ability to configure automated actions for exceptions and reduces risk by minimizing manual intervention
- TPH enables speed of settlement and eliminates unnecessary processing and enables banks to consolidate disparate payment systems and processes into a single standardization solution



TPH lowers the cost of ownership, by increasing efficiency, reducing maintenance costs and its associated risks. Payments hub enables banks to provide their customers a harmonized payment experience by offering 360° view of payments in real time. It enables supervision and centralized control of liquidity, allowing for better cash management for financial institutions and their customers.

Supported Swift Message Types

- Temenos supports the following Swift Message Types:
 - MT101 (Single & Multiple) – Request for transfer
 - MT103 – Single customer credit transfer
 - MT103+ – STP Single customer credit transfer
 - MT191 – Request for payment of charges, interest and other expenses
 - MT195 – Queries
 - MT196 – Answers
 - MT199 – Free format message
 - MT200 – Financial Institution Transfer for its Own Account
 - MT202 – General Financial Institution Transfer
 - MT202C – COV General Financial Institution Transfer
 - MT019 – Abort Notification(used for TARGET2)
 - MT 210 - Notice to Receive
- Clearing specific message types are also supported in Payments Hub(like SEPA,BACS,FEDWIRE,USACH etc.,).



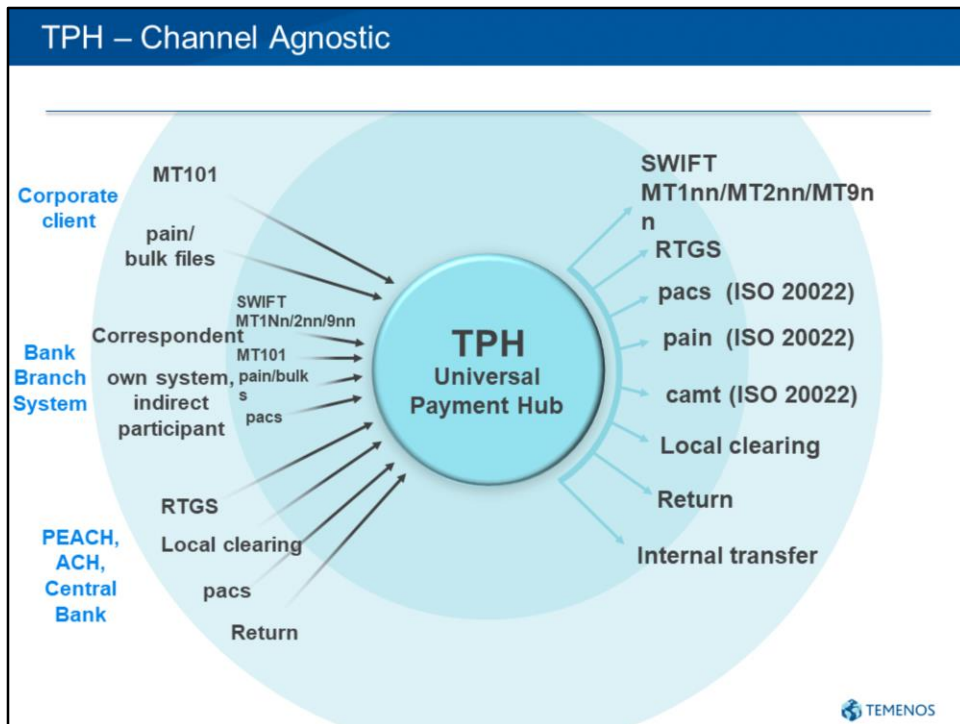
MT195 & MT196 is supported using EB.QUERIES.ANSWERS

MT199 is supported using EB.FREE.MESSAGE

MT200 will be re-directed as MT202

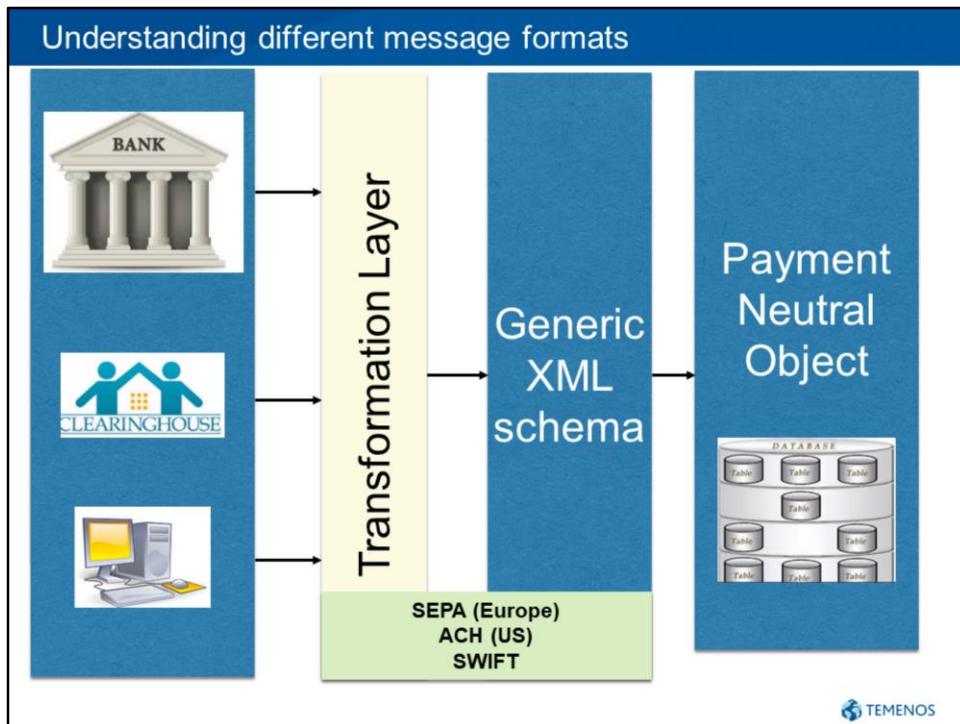
MT210 (Notice to Receive) can be generated and also configured based on bank conditions of account with Institution whether account will be ultimately credited.

RMA check can be performed when MT210 is required to be generated

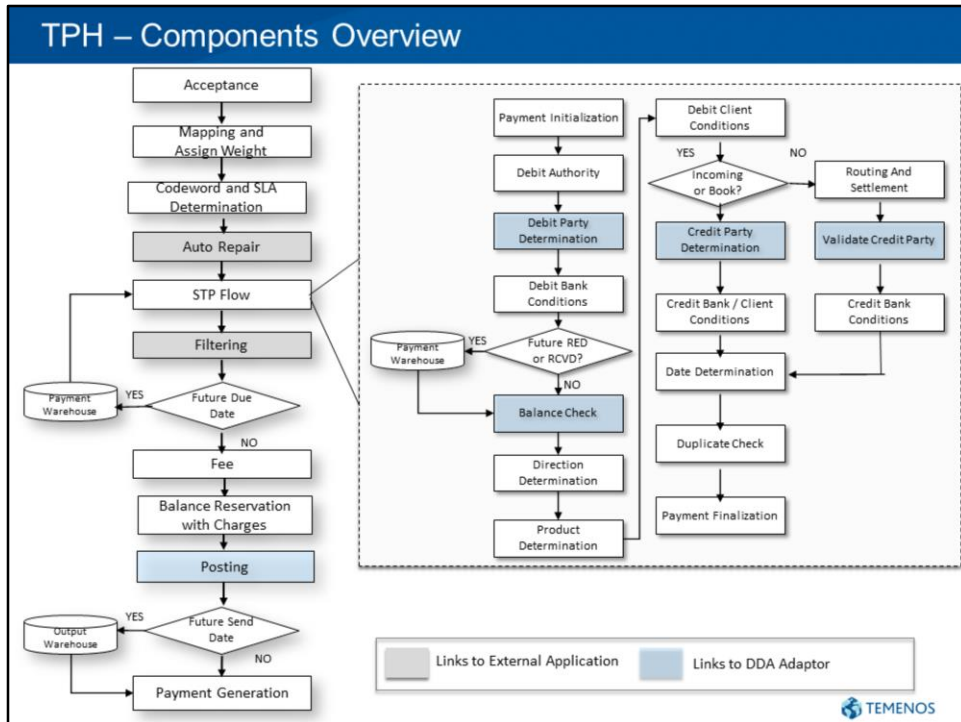


TPH is Payment Hub built on Global payment process which enables Banks / Financial Institutions move to a sophisticated centralized infrastructure that operates Any format, any channel to any destination.

Temenos Payments Hub (TPH) is a universal payments hub solution. It has a Universal Payments Process which enables centralization of all payment operations under a central infrastructure to support multiple payment formats and global payment mechanisms. The payment hub solution provides banks with a single solution to configure their payment processing for all customers, specifying different processing for high and low care customers by parameterization without requiring any software changes. TPH offers excellent breadth and depth of functionality as well as flexibility.



Message Mapping Component interprets the external payment messages from native format(i.e. SWIFT) to Neutral Payment Object in order to map the messages into TPH specific format and to create payment order transaction.



TPH Payment Processing Flow Diagram

Configuration Driven to enable high STP rates

Admin Menu > Payment Hub

Payment Hub	^
Static Data GUI	v
Authorization Principle	⌘
Message Acceptance GUI	v
Payment Router GUI	⌘
Message Mapping GUI	v
Batch Server GUI	⌘
Local Clearing GUI	v
Trip Status Action	⌘
Payment Workflow GUI	v
Weight Assignment	⌘
Inbound CodeWords GUI	v
Outbound CodeWords GUI	v
SLA Per Codeword	⌘
Bank Conditions GUI	⌘
Debit Authority GUI	v
Balance Check GUI	v
Product Determination GUI	v
Client Conditions GUI	v
Routing And Settlement GUI	v
Date Determination GUI	v
Risk Filter GUI	v
Fee Determination GUI	v
Filtering GUI	v
Posting Scheme GUI	v
Claims GUI	⌘
Automated Repair Tool GUI	v
Clearing Directory GUI	v
Payment Purpose Code	v
Payment Categ Purpose	v
SEPA Clearing Directory GUI	v
Templates GUI	v
Inward Entry GUI	v
Non Customer Accounts	v



TPH componentized solution caters to accommodate emerging changes in the market. (i.e. Highly Parametrical Solution). All components put together process a payment transaction as per configuration done.

Dashboards

Pending and Processed Payments

Company	FT Number	Ccy	Amnt	Acc Comp	Acc Num	Acc Ccy	Date	Cust Ref	Sender Ref	Status Code	Description
BNK	BNK181070LFGGFH	EUR	152.00	BNK	EUR1402000010001	EUR	17 APR 2018	13E09T07123321	13D07123D441234	999	Payment complete
BNK	BNK181070HQHDMID	EUR	11.03	BNK	EUR1402000010001	EUR	17 APR 2018	533NDTOEND1013	533DTONND441013	999	Payment complete
BNK	BNK18107FCLJHCCD	EUR	150.00	BNK	EUR1402000010001	EUR	17 APR 2018	123NDT01231123	123D00XN123431321	999	Payment complete
BNK	BNK18107MBJLDFUJ	EUR	11.02	BNK	EUR1402000010001	EUR	17 APR 2018	53ENDTOEND1512	53D0TXND141512	999	Payment complete
BNK	BNK18107LDMFMCMD	EUR	11.01	BNK	EUR1402000010001	EUR	17 APR 2018	5ENDTOENDID1311	5D0T3ND11011	999	Payment complete
BNK	BNK18107LGMJHGC	EUR	149.00	BNK	EUR1402000010001	EUR	17 APR 2018	1END1237708321	1D123N7877123	999	Payment complete

Pending and Processed Enquiry provides details of the processed payments and current status of the payment.

Mid Office Payment Entry Screen

Order Entry Outgoing Transfer (CTR) | OC1819173242

Payment Processing Information

Status	135	Processing Company	BKAC	Processing Date	10 JUL 2018	Processing Date Imposed Flag	☐ Yes
Transaction Reference Number		Priority	1	Product		Output Channel Imposed Flag	☐ Yes
Sender's Reference Number		Transaction Currency		Transaction Amount		Instructed Currency	
Related Reference		Charge Option	☐ Ben ☒ One ☐ Out	Beneficiary ID		Instructed Amount	
Source	OE	Receiver Institution		Receiver Institution H/C/C			
Direction	O						
Transfer Type	C						
Incoming Message Type	RFC						
Balance Reservation							
Balance Reservation Number							
Balance Reservation Debit Chg Acct							
Pre-authorization Number							
Intra Company Payment	☐ Yes						
Share Template Values							
Select from Template							
Save As Template	☐ Yes						
Back Name							

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It is used for Manual payments input by Bank operator i.e. Client initiated payment through Non-STP channels like FAX or to capture the payment transaction of STP failures etc.

Exception Handling - Repair and Resubmit Payments

Repair View Screen OE18107VNZLW

Payment Processing Information	
Status	235
Transaction Reference Number	BNK18107HLLMOLCL
Sender's Reference Number	INCOMINREPAIR
Priority	1
Source	SWIFT
Output Channel	LEDGER
Direction	I
Transaction Currency	USD
Transfer Type	C
Incoming Message Type	103
Balance Reservation	A
Balance Reservation Number	ACLK1810725207
Processing Company	BNK
Processing Date	17 APR 2018
Transaction Amount	999.00
Charge Option	Sha
Sender Institution	BARCGB22

Debit Credit Info **Error Information** Charge Information Routing Information Additional Information Audit

Special Instruction
Error Information
Functional Error.1

CPD10003 ** Credit Main Account: Account Invalid 8 / 1113 /



Payment transactions resulted into repair due to incorrect data or payments validation failure can be repaired using this screen

In detail audit logs – Trace everything that happens to a payment

Audit Trail				
Event Type	Event Description	Error Code	Add Info	History Timestamp
INF	Message mapped according to parameters			17 APR 2018 16:10:48.397
INF	Weight assigned successfully		HLSWF	17 APR 2018 16:10:48.397
INF	Inbound Code/Word processed successfully			17 APR 2018 18:26:34.229
INF	SLA successfully determined		*	17 APR 2018 18:26:34.229
INF	Automated Repair Tool not required for this weight and specific weight		HLSWF	17 APR 2018 18:26:34.229
INF	Payment Successfully Initiated			17 APR 2018 18:26:34.229
INF	Debit Authority successful - Debit Authority Indicator set		S	17 APR 2018 18:26:34.229
INF	Debit Party Successfully Determined		BNK23752USD	17 APR 2018 18:26:34.229
INF	Bank Condition read successfully		DEFAULT.20180417YYY	17 APR 2018 18:26:34.229
INF	Debit Bank Conditions updated successfully			17 APR 2018 18:26:34.229
INF	Own Account successfully Determined		Y	17 APR 2018 18:26:34.229
INF	Direction and Transfer type successfully determined		IC	17 APR 2018 18:26:34.229
INF	Product determination successful		HWAIPHPC14112F78HC.20180417SWFTDEFAULT.EDGEDEFAULT.HKCLG_CODEFAULT	17 APR 2018 18:26:34.229
INF	Credit Party successfully determined		11193 / BNK /	17 APR 2018 18:26:34.229
INF	Debit Main Account Successfully Validated		BNK23752USD	17 APR 2018 18:26:34.229
INF	Credit Main Account Successfully Validated		BNK11193USD	17 APR 2018 18:26:34.229
INF	The Funds Authorization process is skipped for the payment			17 APR 2018 18:26:34.229
INF	Client Condition read successfully		DEFAULTSWFT*****IDEFAULTSWFTY*	17 APR 2018 18:26:34.229
INF	Requested Credit Value date is in past and it cannot be met		20180417/20180719	17 APR 2018 18:26:34.229
INF	Date Determination Successful		Proc:20180417/CVD:20180417/DVD:20180417/PSD:20180417/IBD:20180417/EXD:20180417	17 APR 2018 18:26:34.229
INF	Duplicate check performed successfully			17 APR 2018 18:26:34.229
INF	FATF Regulations Successfully met			17 APR 2018 18:26:34.229
INF	STP successfully completed			17 APR 2018 18:26:34.229
INF	End of STP processing in Single flow service		60331	17 APR 2018 18:26:34.229
INF	Status code changed to ready for Risk Filter			17 APR 2018 18:26:34.229
INF	Risk Filtering successful			17 APR 2018 18:26:34.229
INF	Record key - ClientCharges table used for peeling off		1-20180316YYYBNKYYY	17 APR 2018 18:26:34.229
INF	Fee Processing is successful			17 APR 2018 18:26:34.229
INF	The Balance Check with charges process is skipped for the payment			17 APR 2018 18:26:34.229
INF	Posting lines and statement lines generated successfully and passed to Ledger			17 APR 2018 18:26:34.229
INF	Posting completed successfully		BNK18107GCGGFCM	17 APR 2018 18:26:34.229
INF	Payment processing is complete			17 APR 2018 18:26:34.229



Audit Trail option provides the details of transaction processed by respective components and its' outcome.

Introduction to concept of POR tables



Payment Transaction related data capture table known as “POR” Tables which are updated by TPH during the processing considering the Static Data maintenance and Received Message data or User Input.

- POR Table (Transaction details captured by the system)

POR tables (payment order tables)

POR.TRANSACTION

POR.SUPPLEMENTARY.INFO

POR.POSTING.AND.CONFIRMATION

POR.AGREEMENT.AND.ADVICE

POR.AUDIT.TRIAL

Business Case 1



Process incoming MT103

Process incoming MT103

Alfa Beta in US, wishes to pay USD 340 to his supplier Nike located in UK.

Nike holds an USD account with T24 Bank while Alfa Beta has an USD account with CITI Bank in US, NY

CITI Bank, on behalf of Alfa Beta sends a MT103 to T24 Bank

CITI Bank shares a direct account relationship with T24 Bank.

T24BANK uses TPH (Temenos Payment Hub) for handling all types of payment transactions and T24 acts as the DDA system (Demand Deposit Account)

Transaction charges on the receiver's side is NIL

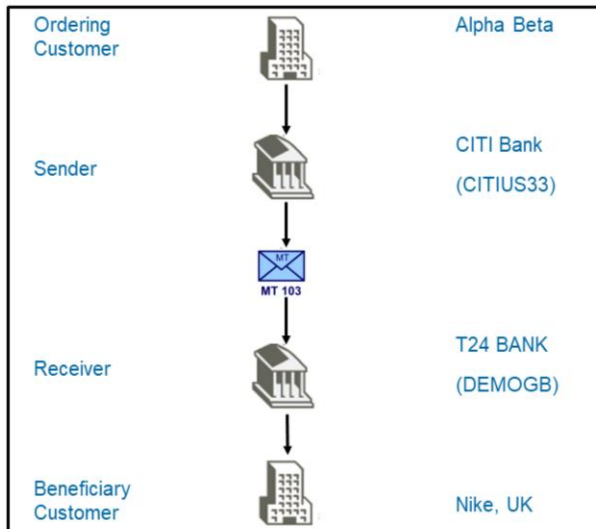
Charge type used is SHA



T24 Bank – Account with Institution / Beneficiary Customer

TPH – Payment Processing company / Branch

Process incoming MT103



Flow of MT 103 message.

Process incoming MT103

```
SWIFTMT103-BUC1-001 (2)txt
1 |1:F01DEM0GBFXAXXX9024629991|2:01031425131119CITIUS33AXXX892459501313111914355|3:{108:BNKMT103FXSHA}}4:
2 :20:BUC1MT013INC
3 :23B:CRED
4 :32A:160719USD340,00
5 :50K:/GB10MIDL40051574128754
6 Alfa Beta
7 New York
8 US
9 :52A:MIDLGB22XXX
10 :59:/11193
11 Nike
12 London
13 UK
14 :70:INCOMING STP NO FX
15 :71A:SHA
16 ~}
```



SWIFTMT103-BUC1-001.txt

Incoming message received from SWIFT channel

Introduction to payment neutral object

Received File Details Received Bulk Details Received Message Details

Received Message/File Details - List

File Reference	Queue Name	Company	File Status	File Name	Source	Single Multiple Ind
ABC	SWIFT		RECEIVED			

Received File Details Received Bulk Details Received Message Details

Received Message/File Details - List

File Reference	Queue Name	Company	File Status	File Name	Source	Single Multiple Ind
ABC	SWIFT	BNK	MAPPED		SWIFT	S

Blob Payment File Content

Blob Unique ID Message Content

ABC

(1F01DEM0GBFXA000024629991)(2:01031425131119CITRUS33AXXX08924595013111914355)(3:108BNKMT103FXSHA)(4:20BUC1MT013INC:23B CRED:32A160719US0340,00:50K/GB10MDL42051574128754:Alfa Beta:New York:US:52A:MDLGBZ2XXX:59/111193:Nike:London:UK:70:INCOMING STP NO FX:71A:SHA:3)

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Using menu option “Payment Hub -> Received Files” Message status and received message details can be viewed.

Introduction to concept of POR tables

● POR.TRANSACTION main payment object table

Pending and Processed Payments

Company	FT Number	Ccy	Amt	Acc Comp	Acc Num	Acc Ccy	Date	Cust Ref	Sender Ref	Status Code	Description
BNK	BNK18107GFCGFCM	USD	340.00	BNK	11193	USD	17 APR 2018		BUC1MT013INC	999	Payment complete

POR TRANSACTION BNK18107GFCGFCM

CompanyID	BNK	Debit Main Account	23752	Credit Client ID	10043
FTNumber	BNK18107GFCGFCM	Debit Main Account Currency Code	USD	Credit Business Line	999
Status Code	999	Debit Party Line1	CDC Group Inc. (New York	TransactionAmount	340.00
Payment Direction	I	Debit Party Line2	399 Park Ave	TransactionCurrencyCode	USD
C T R B T B Indicator	C	Debit Party Line3	New York	Transaction Amount Home	340.00
Own Account Indicator	Y	Debit Main Amount	340.00	Home Currency	USD
Status Date Time	20180417102634279	Debit Main Amount Home	340.00	Transaction Amount C.L Home	340.00
Specific Weight Code	SWF	Debit Client I D	100492	C.L Home Currency	USD
Weight Code	N	Debit Business Line	999	Details Of Charges	SWF
Heavy Weight Product I D	PPHPC14112796C.20180101	Credit Main Account Company I D	BNK	Sold Value Date	17 APR 2018
Message Priority	1	Credit Main Account	11193	Credit Value Date	17 APR 2018
Domestic International	D	Credit Main Account Currency Code	USD	Originating Source	SWFT
MessageFormat	103	Credit Party Line1	N/A	Received Source	SWFT
Priority Code	S	Credit Party Line2	1 BOWERMAN DRIVE	Originating Channel	SWFT
Company Region	ALL	Credit Party Line3	BEAVERTON	Source Product	SWFT
Sender Address	OTFV33	Credit Main Amount	340.00	Incoming Message Type	103
Receiver Address	DSM00PK	Credit Main Amount Home	340.00	Output Channel	103038
Senders Reference Incoming	BUC1MT013INC	Credit Charge Exchange Rate	1.233668000	Single Multiple Indicator	S
Bank Operation Code	CRD	Credit Charge Currency Pair	GBP/USD	Number Of Children	0
Debit Main Account Company I D	BNK	Credit I B A N Indicator	N	Message User Reference	SWFT103038A



POR.TRANSACTION captures payment transaction details processed by TPH along with current status of the payment.

Introduction to concept of POR tables-POR.SUPPLEMENTARY.INFO

POR.SUPPLEMENTARY.INFO BNK18107FCGGFCM Please Select 18-03

Main Or Charge Account Type		Account Company Id	Account Number
D	BNK	23752	
C	BNK	11193	

Additional Information Code	Add Inf Type Line Seq	Additional Inf Tag	Additional Inf Line
RMTRNF	1	70	INCOMING STP NO FX

Party Type	Party Role	Role Indicator	Information Tag
CREDIT	BENFCY	R	59
DEBIT	SENDER	R	
DEBIT	ORIGPTY	R	50K
DEBIT	ORDINS	R	50A

Entry ID#	Value
Entry ID# 1	184344711799403.00
Entry ID# 2	1-2
Entry ID# 3	1
Ed Duplicate Type Id	PH-INCOMING

Ccy	Account Type	Account Status
USD	N	Active
USD	C	Active

Reserved6	Reserved7

Party Identifier Code	Party Account Line
	/11193
CITRUS33	/GB10MDL40051574128754
MDLGE22	

Introduction to concept of POR tables- POR.POSTING.AND.CONFIRMATION

POR.POSTING.AND.CONFIRMATION
BNK18107GFCGFCM

Please Select

Charge Party Indicator	Fee Type	Charge Sign
C	TRANSACTIONFEE	P

Booking Date	17 APR 2018	17 APR 2018
Transaction Ref	BNK18107GFCGFCM	
Local Currency Code	USD	

Posting Line Number	Statement Line Number	
1	1	Cust CHDebit HKCL with charges
2	1	Bank HKCLG standard
3	1	Charge Credit Detailed

Type Of Charge	Fee Description	Ch
R	Transaction Fee	6.18

Booking Date	17 APR 2018	17 APR 2018
Transaction Ref	BNK18107GFCGFCM	
Local Currency Code	USD	

Statement Line
Cust CHDebit HKCL with charges
Bank HKCLG standard
Charge Credit Detailed

Introduction to concept of POR tables- POR.AGREEMENT.AND.ADVICE

POR.AGREEMENT.AND.ADVICE			BNK18107GFCGGFCM			- Please Select		
Debit Credit Indicator			Client Conditions Id					
C			DEFAULTSWIFT*****					
<			>					
Debit Cdt Bank Cond Ind			Bank Conditions Id					
D			DEFAULT-20160101					
<			>					

Pc Language Id		Pc Statement Format Name		Billing Indicator	
1				N	
<				>	
Cur Non Stp Indicator		Btr Non Stp Indicator			
N		N			
<				>	

Charge Posting Separately		Charge Posting Detail		Vat Principal	
N		N		0	
<				>	
Warehouse Flag		Warehouse Release Time		Charge Account Company Id	
J					
<				>	

Vat On Charge		Non Stp Indicator		Fx Non Stp Indicator	
N		N		N	
<				>	

Party Roles

Party Role List

PartyRole	Party Role Tag	Description	
ACWINS	57	Account with Institution	∞ / ∅
ASVINS	52	Account Servicing Institution	∞ / ∅
BENFCY	59	BENFCY	∞ / ∅
BENINS	58	Beneficiary Institution	∞ / ∅
IMPCDT	50	Implied Credit	∞ / ∅
IMPDBT	50	Implied Debit Account	∞ / ∅
INSPTY	50	Instructing Party	∞ / ∅
INTINS	56	Intermediary Institution	∞ / ∅
ORDINS	52	Ordering Institution	∞ / ∅
ORDPTY	50	Ordering Party	∞ / ∅
RCVCBK	54	Receiver's Correspondent Bank	∞ / ∅
SENDER	50	Sender	∞ / ∅
SNDCBK	53	Sender's Correspondent Bank	∞ / ∅
SNDINS	51	Sending Institution	∞ / ∅
TRMINS	55	Third Reimbursement Institution	∞ / ∅
ULTCDT	99	Ultimate Creditor	∞ / ∅
ULTDBT	99	Ultimate Debtor	∞ / ∅



Quick Reference Guide for party roles.

Contains party role name, description and tag from which value will be retrieved for the party

INSPTY

Instructing Party

50 C or L

ORDPTY

Ordering Party

50 A, F, G, H or K

SNDINS

Sending Institution

51A

ASVINS

Account Servicing Institution

52A or C

ORDINS

Ordering Institution

52A or D

SNDCBK

Sender's Correspondent Bank

53A, B or D

RCVCBK

TRMINS

Third Reimbursement Institution

55A, B or D

SENDER

Sender

IMPDBT

Implied Debit Account

INTINS

Intermediary Institution

56 A, C or D

ACWINS

Account with Institution

57 A, B, C or D

BENINS

Beneficiary Institution

58 A or D

BENFCY

Beneficiary

59A or no letter option

IMPCDT

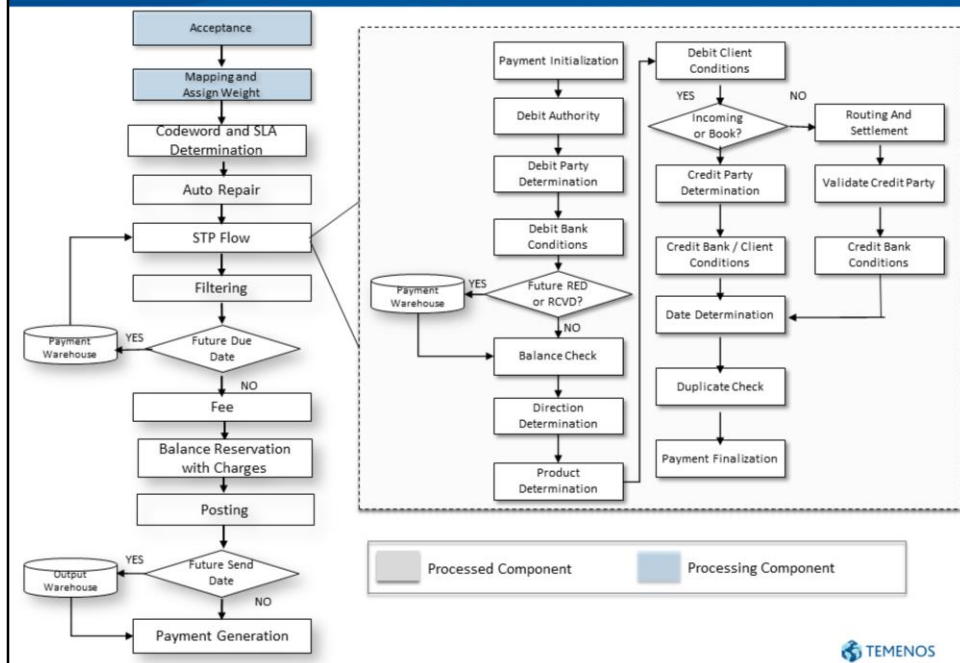
Implied Credit Account

RECVER

Receiver Institution of the MT103 message

Receiver's Correspondent Bank
54A, B or D

How does TPH process a payment message?



For a payment to be processed, number of internal activities need to happen on the payment. The activities are listed and each of them are impact payment processing in a specific way. This is explained in detail as we proceed.

Message Receiving and Mapping

Message Format	Support
SWIFT	Consumed as is by TPH. No transformation required.
SEPA	Native formats for SEPA messages are supported.
US ACH	Received by the US regional layer, de-bulked and sent to TPH in a bilaterally agreed format. Communication between the Regional layer and TPH is in an embedded mode.
TARGET2	Native Formats are supported.
Messages from Payment Order	Directly mapped to TPH internal tables. Communication between Payment Order and TPH is in an embedded mode.
Proprietary format messages	To be mapped to the exposed XML schema



As a payments hub, TPH can receive single as well as bulk messages. Messages once received are stored in the database so that no message is lost. Once the message is accepted and successfully validated, TPH performs what is called 'mapping' which maps the data received on to various internal TPH tables which ease payments processing.

Being a payments hub, there would be a number of different message formats that need to be consumed. In order to cater to this, TPH exposes a XML schema to accept credit transfer as well as direct debit messages.

Hence, once the messages are mapped to this XML schema and sent, TPH, on successful validation will store the message as well as map them. The XML schema can accept both single as well as bulk messages.

Concept of Message Acceptance and Mapping

Tables Used	
F.PRF.BLOB	To store the received message
F.PPT.RECEIVEDFILED ETAILS	Stores the attributes of the received message such as source, format etc.

Messaging process consists of below stages:

- RECEIVED: Original Message is received
- ACCEPTED: Post successful validation of received message
- MAPPED: Transformed into TPH and status assigned for payment transaction process



What can TPH do with messages that it receives?

Receive message from any Channel - Channel agnostic

As the name implies, Message Acceptance denotes accepting messages from any channel.

Can receive single or bulk messages

Messages can be single messages or bulk messages.

Interpret/Validate messages

Messages such as SWIFT need to be interpreted to obtain message formats whereas messages such as Batch messages need to be validated for content. Hence, both interpretation and validation of messages are performed when a message is accepted.

Accept or reject messages

Once a message is received, basic message validation such as to check if it is supported message type, supported channel etc need to be performed. Based on validations set at the channel level/message level, message is either accepted for further processing or rejected.

Mark messages to be repaired and resubmitted

If messages result in an error, facility to resubmit the message is available.

Forward messages

There could be messages that are received by TPH which aren't meant for processing within TPH, such messages can be identified and forwarded to a pre-determined queue. (This is only provided for messages in 1 and 2 series). Banks are expected to route the correct messages to TPH.

Check for technical duplicates

In order to avoid messages getting processed more than once, duplicate processing is enabled at TPH level for bulk messages.

Send ACK/NACK responses

Once a message is received, ACK (Acknowledgement) or NACK (Negative acknowledgement) can be sent to a pre-configured queue

Transform messages to neutral format (Mapping)

Messages received in any format can be transformed (mapped) into a Payment Engine generic internal format so that TPH can perform further processing. The message is parsed into individual fields.

Create Payment Order

Based on the parsed message, a payment order is created and stored in the database.

Messages can be accepted 24/7

Messages can be accepted from any channel at any time – even while SODEOD process is in progress in T24

Messages can be mapped only when TPH is online

Messages can be mapped only when the system is online (Not while SODEOD is in progress)

Concept of Message Acceptance and Mapping

Received Message/File Details - List						
File Reference	Queue Name	Company	File Status	File Name	Source	
TPS18016000016003	EBAINST	BNK	MAPPED	INSTANT-PACS008-TPS18016000016-003.xml	EBAINST	
TPS18016000016001	TPHINSTC2B		MAPPED	INSTANT-PAIN001-TPS18016000016-001.xml	CTINST	
TPS18016000016004	EBAINST	BNK	MAPPED	INSTANT-PACS008-TPS18016000016-004.xml	EBAINST	
TPS18016000016005	TPHINSTC2B		MAPPED	INSTANT-PAIN001-TPS18016000016-005.xml	CTINST	
TPS18016000016006	EBAINST	BNK	MAPPED	INSTANT-PACS008-TPS18016000016-006.xml	EBAINST	
TPS18016000016007	EBAINST	BNK	MAPPED	INSTANT-PACS008-TPS18016000016-007.xml	EBAINST	
TPS18016000016008	EBAINST	BNK	MAPPED	INSTANT-PACS008-TPS18016000016-008.xml	EBAINST	
TPS18016000016009	EBAINST	BNK	MAPPED	INSTANT-PACS002-TPS18016000016-009.xml	EBAINST	
TPS18016000016010	EBAINST	BNK	REJECTED	INSTANT-PACS002-TPS18016000016-010.xml	EBAINST	
TPS18016000016013	EBAINST	BNK	MAPPED	INSTANT-PACS002-TPS18016000016-013.xml	EBAINST	
TPS18074880001002	CHQCLGXML	BNK	MAPPED	SEPA-PACS003-TPS18074880001-002.xml	CHQCLG	
TPS18017880001011	TPHDDI		ACCEPTED	SEPA-PAIN008-TPS18017880001-011.xml	DDI	
TPS18074880001003	CHQCLGXML	BNK	MAPPED	SEPA-PACS003-TPS18074880001-003.xml	CHQCLG	
TPS17311000016004	STEP2	BNK	MAPPED	SEPA-PACS003-TPS17311000016-004.xml	STEP2	
TPS17311000016007	TPHCTI		MAPPED	SEPA-PAIN001-TPS17311000016-007.xml	CTI	
TPS17311000016002	STEP2	BNK	MAPPED	SEPA-PACS008-TPS17311000016-002.xml	STEP2	
SWIFT-MT101-TPS16173000001-005	SWIFT	BNK	MAPPED		SWIFT	
SWIFT-MT103-TPS16173000001-011	SWIFT	BNK	MAPPED		SWIFT	
SWIFT-MT103-TPS16173000001-009	SWIFT	BNK	MAPPED		SWIFT	



How to view the status of a message once it has been posted to TPH

Using menu option "Payment Hub -> Received Files" Message status and received message details can be viewed.

Are there different statuses for a message?

Messages are posted on to Queues and these messages are picked up by TPH for processing. At the message acceptance level, use the following set of menus to view the status of messages/Resubmit messages. Services need to be running in the background for the messages to be picked up and processed by TPH.

Status of message at this juncture can be any one of the following

RECEIVED

Message has been received by TPH. No validation or any sort of processing has commenced on the message.

ACCEPTED

Message is in the supported format, is from a valid channel and has been accepted for further processing

MAPPED

An accepted message of TPH has been mapped to a neutral format and a payment order has been created. Financial processing can commence on the message.

REJECTED

Message is not a supported format/incorrect message. If a message is rejected by

TPH, then, the only way would be to make the corrective action (If any) and re-send the message to TPH.

REPAIR

Message is a supported format but is an erroneous message. A message in REPAIR can be resubmitted post corrective action. Message need not be re-pumped in.

FORWARDED

Message has been forwarded to another queue.

Message Acceptance - Configuration

Admin Menu > Payment Hub > Message Acceptance GUI > Message Acceptance Param

ID	SWIFT
Field Name	Field Value
Incoming Message Directory	
Channel	SWIFT
Single Multiple Indicator	- S - B - C
Interpret API	SwiftService.interpretSwift
Validate API	
Check Dup Ind	- Y - N
ACK Required Y?	- Y - N
ACK API	
ACK NACK Queue	
Message Forward API	SwiftService.forwardSwiftMessage
Message Forward Queue	
Read Message API	SWIFT.IN
Debulk API	



Define the message formats to be used

For TPH to be able to pick up and process messages, it needs to know the valid message formats that it can process else it would have to reject them.

All recognized message formats need to be defined.

The key is the Message Format and inside it contains a descriptive text of the message format.

Any message which has a format other than the ones listed as part of this table will be rejected by TPH

Any incoming message will have the message format as part of the message.

Define which message formats can come via which channel

While there can be numerous message formats and number of channels, certain message formats can be dealt only by certain channels. Hence, message formats need to be linked to channels for ease of processing.

Combination of Channel name, Message Format and Direction of the message identify each record uniquely.

Message Direction indicates the direction of the message

R – Receive (Incoming)

S – Send (Outgoing)

Define the rules for accepting a message

Message Acceptance Param(eter), as the name suggests, defines the various rules

while processing messages from a specific queue and hence the key is the queue name.

Various types of messages would come into TPH. For instance,

- It can be message to process salaries for all employees is received. Such a message would be a bulk message as there would be multiple credits and one debit.
- It can be a single SWIFT message (For instance 101)
- It can be a single credit message from Payment Router channel
- It can be a BACS local clearing file

How will TPH know

Whether it is a single or a bulk message?

Which channel these types of messages can originate from?

Whether an acknowledgement needs to be sent once the message has been accepted or a negative acknowledgement needs to be sent if the message is incorrect (ACK/NACK)?

If ACK/NACK needs to be sent, where should such return messages be placed?

Should any API be triggered to validate/interpret the message?

Should technical duplicates be checked for?

All such details are defined in here.

Key to the records here is always a channel from which the message comes in.

Message Mapping - Configuration

Admin Menu > Payment Hub > Message Mapping GUI > Message Mapping Param

Payment Hub

Static Data GUI

Authorization Principle

Message Acceptance GUI

Message Acceptance Parameters

Message Formats

Message Format Per Channel

Modify

Audit

Debulk API

Mapping API

Source Type API

Clearing Transaction Type

Clearing Nature Code.1

HKCLGMessageMappingService.debulkHKCLGMessage

HKCLGMessageMappingService.doMessageMappingHKCLG

- Please Select



All valid message types need to be defined in PP.MSGPAYMENTTYPE

While there can be numerous message types and number of channels, certain message types can be dealt only by certain channels. Hence, message types need to be linked to channels for ease of processing. This is defined in PP.MSGPAYMENTTYCHANNEL

Once a message has been accepted by TPH for processed, it needs to transform the message into a neutral format so that further payment processing can happen. Mapping phase is the one that defines how the transformation needs to happen. No financial calculations/processing happens at this stage.

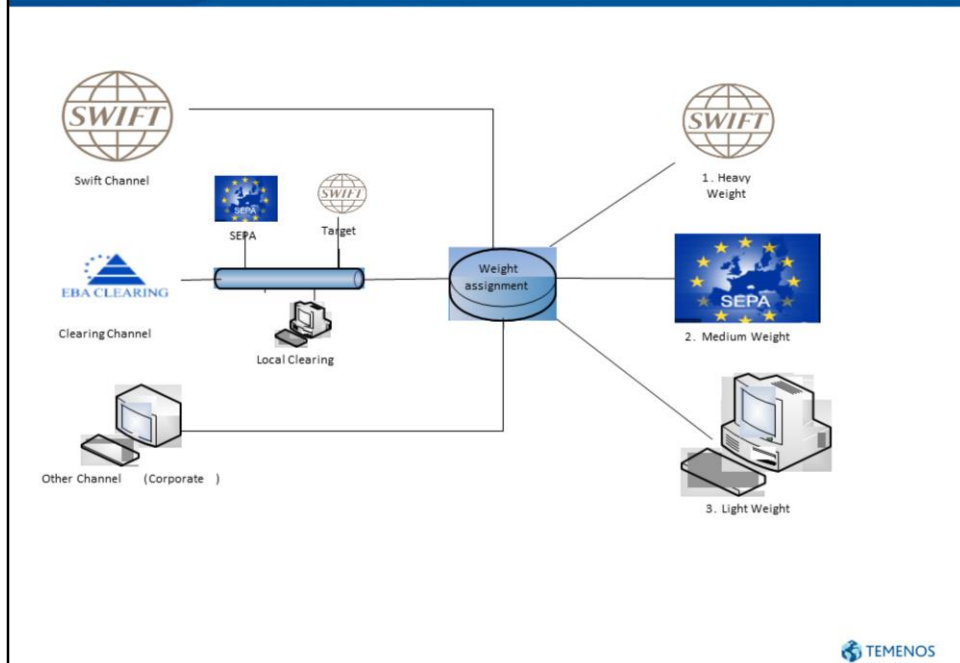
Mapping API

SourceType API

*API which can be used perform changes to the message based on payment values.
Optional.*

For instance, the API shown above will distinguish if the incoming message is a CHAPS or FPS based on the sender's BIC.

Assign Weight for the Payment



A global payment engine receives a number of payments throughout the day and has to process all of them as quickly and efficiently as possible. Hence, once a message is accepted, before any financial processing can commence on the message, it is vital to check if there are characteristics in the Payment Message that will influence payment processing.

This weight refers to an overall classification of the payment on a very broad level. Weight can be defined as a categorisation of a payment instruction with a purpose to influence flavour of processing.

The more complex the payment attributes are and more combinations they can have, the higher the weight they need to possess.

SWIFT - Heavy weight

Outgoing messages in SEPA clearing/USACH clearing – Medium weight

Other local clearing payment messages – Light weight

Assign Weight for the Payment

Admin Menu > Payment Hub > Weight Assignment

Payment Hub	^
Static Data GUI	▼
Authorization Principle	⌕
Message Acceptance GUI	▼
Payment Router GUI	⌕
Message Mapping GUI	▼
Batch Server GUI	⌕
Local Clearing GUI	▼
Trip Status Action	⌕
Payment Workflow GUI	▼
Weight Assignment	⌕

ID	STEP2.pacs.008.CT-20170203
Field Name	Field Value
Company ID	BNK
Rank	12
Specific WeightCode	SCT
Weight Description	Incoming SEPA Credit Transfer
Weight Code	- H - A - M
Start Date	3-Feb-17
End Date	DD MM YYYY
AutoRepair Instance Name	

Company ID	Originating Source	Message Type	Rank	Specific Weight	High Level Weight
BNK	*	*	3	P2	H
BNK	SWIFT	103	1	P1	H
BNK	SWIFT	*	2	P3	H



Can there be specific weights based on attributes of a message?

Yes, this is possible.

Most payments can be grouped under one of the 3 high level weights. But there could be expectation for special treatment for smaller sets of payments which is not covered directly by the high level weights.

Specific weight can be understood as the categorisation through which customised treatment of those payments can be provided. However, all specific weights have to relate to a high level weight. Specific weight is specialisation of high level weight.

In order to understand how weights are assigned to messages, let's use the table shown above.

Rank dictates the which record will be used if multiple records are chosen (Rank 1 will have the highest priority)

The '*' in the above table means that the value in that place can be wild-carded. This can be used when a bank may want to assign a particular weight to all the messages from a particular Originating Source or all the sources for a particular Message Type.

Once a weight is assigned, can it be changed in downstream processing?

No. This is not possible.

Can a message not have a weight defined?

No. Every message needs to have a weight defined. Else, system will throw an error and processing will stop.

Skip Programs based on weight

Admin Menu > Payment Hub > Static Data GUI > ProgramsPerWeight

ProductDetermination.H	BNK	*	10	  
ProductDetermination.L	BNK	CDR	4	  
		STL	9	
		ICD	10	
		CCR	15	
		*	20	
ProductDetermination.M	BNK	CQE	28	  
		CQS	29	
		*	30	
RiskFilter.H	BNK	CLM	5	  
		*	20	
RiskFilter.L	BNK	CDR	4	  
		STL	9	
		ICD	10	
		CCR	20	
		*	40	
RiskFilter.M	BNK	CQE	28	  

Flexibility to make a decision whether a specific processing within the payment engine needs to be invoked for a payment leading to reduction in processing time. Some processing may not be applicable to certain payments. This distinction can be done based on weight. Table to be used to configure this is PPL.PROGRAMSPERWEIGHT.

Skip Programs based on weight

ProgramsPerWeight RiskFilter.H

Amend Audit

Company ID: BNK

Specific Weight Code: 1: CLM

Ranking: 1: 5.00

Program Name: 1: RiskFilterService.doRiskFiltering

ProgramsPerWeight Description: 1: EN Skip program

Program Skipindicator: 1: ☒ Y ☐ N

Specific Weight Code: 2: *

Ranking: 2: 20.00

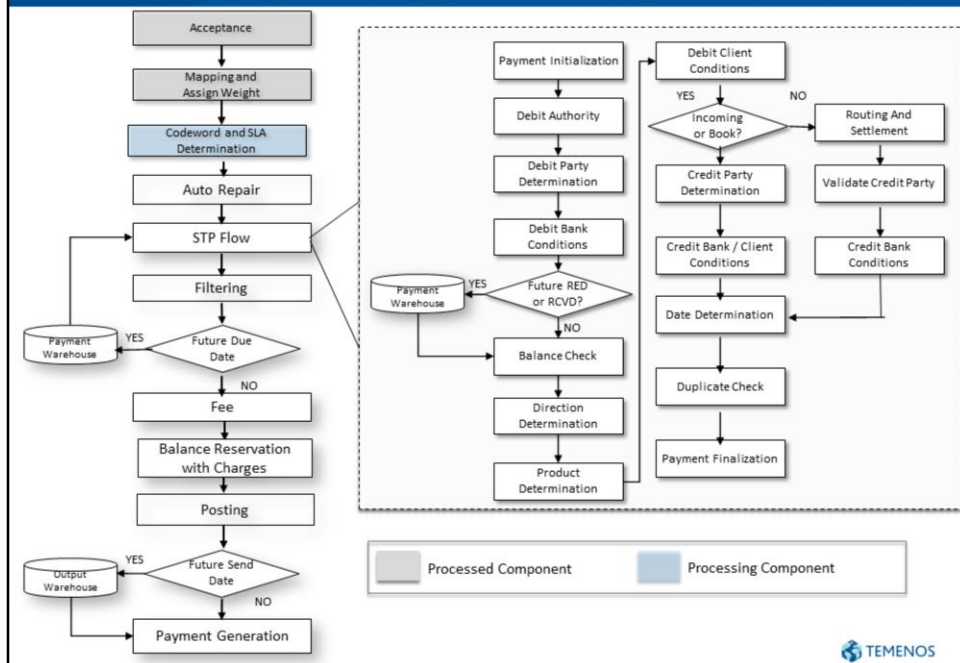
Program Name: 2: RiskFilterService.doRiskFiltering

ProgramsPerWeight Description: 2: EN Process Program

Program Skipindicator: 2: ☐ Y ☒ N

A specific weight code "CLM" is created for CLAIM functionality which is an internal process for creation of MT191 message by creating new transaction as part of payment transaction demands to send charge claim. This type of transaction need not to process all payment programs hence irrelevant programs are skipped using this functionality

How does TPH process a payment message?



For a payment to be processed, number of internal activities need to happen on the payment. The activities are listed and each of them are impact payment processing in a specific way. This is explained in detail as we proceed.

Code Words

Admin Menu > Payment Hub > Inbound Codeword GUI > Inbound Codeword

- Payment Hub ^
- Static Data GUI v
- Authorization Principle ㉸
- Message Acceptance GUI v
- Payment Router GUI ㉸
- Message Mapping GUI v
- Batch Server GUI ㉸
- Local Clearing GUI v
- Trip Status Action ㉸
- Payment Workflow GUI v
- Weight Assignment ㉸
- Inbound Codewords GUI ^
- Inbound Codeword ㉸
- Processing Sequence ㉸

Inbound Codeword | CORT.INSBNK-20160101

Modify
Audit

Company	BNK
CodeWord *	CORT
Information Code *	☐ Timind ☒ Insbrk ☐ Insdr ☐ Regrep ☐ Inschr
Message Payment Type *	+
Originating Source *	+
Codeword Ranking.1 *	10.00
Codeword Text.1 *	+
Codeword Priority for PD.1 *	1
Adjusted Message Priority.1	- Please Select
Processing Sequence Number.1	v
Non STP Indicator.1	☒ Y ☐ N
Fee Codeword Flag.1	☐ Y ☒ N
Outbound Codeword Applicable Flag.1	☒ Y ☐ N
LinkID	CORTINSBNK**-20160101
Start Date	01 JAN 2016
End Date	31 DEC 2099

Codewords are added as part of the payment message to convey payment processing information for financial institutions. ‘*SWIFT codewords*’ are codes wherein SWIFT has defined and detailed the meaning of each codeword along with the method to read and interpret these codes. This is to ensure that various financial institutions process these codes in a consistent manner.

Banks / financial institutions can also make special agreements and come up with codewords to trigger bilateral agreements amongst themselves, the presence of these codewords can trigger special payment processing as agreed between the banks. These are termed as ‘*Bilaterally agreed codewords*’.

Note!

The use of Codewords to identify specific processing requirements is not limited to SWIFT messages. Any other message sources may also supply codewords as part of the payment request.

Codewords received in the payment object (inbound codewords) are searched against the Inbound Codeword table to see if any processing logic is defined for that particular codeword and thereby evaluates, whether the codeword influences / affects the manner in which the payment is to be processed within the payment engine.

Inbound codewords are codewords which are received in the payment instruction as part of the SWIFT fields. Below are the tags where codewords may be present in a SWIFT message.

13C, 23E, 72, 77B and 56/57/58 for //RT

These are read by the Mapper and stored in a payment object. These codewords are then input for Inbound Codeword processing module. The input / inbound codewords are searched in the Inbound Codeword table to see if a match for that particular codeword is available.

If a match is *not* found, then for Inbound Codeword Processing, TPH ignores the codeword in the payment object and does not act on it. i.e. the codeword will not be considered for payment processing

Configure the output results based on the received inbound codewords here. The output results impact the overall payment processing. If a payment has more than one inbound code word, for each code word this table is looked up iteratively. At last, only one inbound code word is selected for product determination based on CodeWordPriorityForPD.

Once a record is selected, applicable output results are considered for payment processing. For example in the above screenshot, the codeword will impact the message priority as specified under 'Adjusted Message Priority'. The code word does not influence Non STP indicator and conditional fees. The code word is also not considered for Outbound codeword processing. Here there is no mentioning about 'Processing sequence number' which ranges from number 1 to 5.

Product configuration is set up to map Codeword INS, BNF and ACC (including code word text) received in field 72 (INSSDR) to map same into the outgoing message (outbound CodeWord Applicable Flag) according SWIFT usage rules.

Code Words – Processing Sequence

Admin Menu > Payment Hub > Inbound Codewords GUI > Processing Sequence

Payment Hub	ID	1
Static Data GUI	Field Name	Field Value
Authorization Principle	Company ID	BNK
Message Acceptance GUI	Processing Sequence Desc	Inbound Processing Sequence 1
Payment Router GUI	Processing Sequence Routine Name	InboundCodeWordService.se tCreditValueDate
Message Mapping GUI	Inbound Processing Sequence Flag	• Y • N
Batch Server GUI	Outbound Processing Sequence Flag	• Y • N
Local Clearing GUI		
Trip Status Action		
Payment Workflow GUI		
Weight Assignment		
Inbound CodeWords GUI		
Inbound Codeword		
Processing Sequence		



Model Bank > PAYMENTS > TPH > Parameters > Inbound Codewords > Processing Sequence

Processing Sequence is a set of pre-defined steps (methods) that influences the payment processing by setting various flags and indicators that are subsequently read and taken into consideration for payment processing by the impacted modules within the payment engine.

If there are multiple code words, Processing Sequences (If any) for all of them are executed and only then is the final code word arrived at.

Detail

Codewords can stimulate special processing rules for each payment, these are then defined as Processing Sequence. Processing Sequence detail out the set of pre-defined processing steps that result in certain flags and indicators being set in the payment object. These flags/ indicators are subsequently read by the impacted payment engine module which in turn influences the processing of the payment within the payment engine.

Processing Sequence related details are stored in a separate database table.

Processing sequence will always be referred to with a number.

Processing Sequence – Additional Information

Processing Sequence	Information Code and Code Words	Result of Processing Sequence API	Flags/Code Words that will be set
1	OTHR/BBDD	Assign Requested Credit Value Date equal to Requested Execution Date	Set Credit value Date Imposed Flag
2	INSBNK (In tag 23E) OTHR/NBALANCE/Dddmmy and /or /Cddmmy	Assign Requested Credit Value date = Credit Value date in the message for Cddmmy and impose the same for processing Assign Debit Value date = Debit value date in the message for Dddmmy and impose the same for processing	- Set the Skip Balance Check flag - Set the SkipBoundaryCheck flag - Set the value of the Skip Fee flag to 'A' (Skip All charges) - Set the value of the Debit Authority Indicator field to 'S' - Set Credit value Date Imposed Flag - Set the Skip Balance Check flag - Set the SkipBoundaryCheck flag - Set the value of the Skip Fee flag to 'A' (Skip All charges) - Set the value of the Debit Authority Indicator field to 'S'
3	INSSDR (In tag 72) REC/NBALANCE	Debit Value Date = Requested Credit value Date in the message	- Set the Skip Boundary Check flag - Set the value of the Skip Fee flag to 'A' (Skip All charges) - Set the value of the Debit Authority Indicator field to 'S'
6	INSBNK (In tag 23E)	Set RED to RCVD if RED is in future and make RED as blank.	

SWIFTgpi: TPH is able to receive field 111 (Service Type Identifier) and field 121 (Unique End-to-End Transaction Reference (UETR)) in header block 3 from SWIFT. The two SWIFT fields are mapped by TPH into the payment order in field 'Service Type Identifier' and 'End To End Reference'. TPH does not forward the information

SLA Determination

Admin Menu > Payment Hub > SLA Per Codeword

Payment Hub
Static Data GUI
Authorization Principle
Message Acceptance GUI
Payment Router GUI
Message Mapping GUI
Batch Server GUI
Local Clearing GUI
Trip Status Action
Payment Workflow GUI
Weight Assignment
Inbound CodeWords GUI
Outbound CodeWords GUI
SLA Per Codeword

ID	BNK-20160101
Field Name	Field Value
Message Priority.1	*
CodeWord.1	*
CodeWord Tag.1	*
CodeWord Text.1	*
Ranking.1	4
SLAID.1	*
Start Date	1-Jan-16
End Date	DD MM YYYY

Service Level Agreement (SLA) is an agreement that a bank provides to its correspondent banks defining the various levels of service in payments processing that it would offer. The SLA will influence the process of payment in payment engine such as applying special bank conditions and Routing and settling the payment in a different manner.

When correspondent bank wishes to use an agreement for a particular message, then in the payment message, special code words agreed for this purpose should be mentioned. The SLA determination component will determine the SLA for the payment based on the SLA configuration with respect to the code words specified in the message.

The output of the SLA determination can be used as a key for further payment processing (like Bank conditions and Routing and settlement)

Overview

The Service level agreement with the counterparty banks applies a special behaviour of a message in the payment engine. The agreement can be a general agreement with the bank or specific level agreements which are linked to specific bi-laterally agreed code words. The code words can be a standard swift code words or bilaterally agreed code words

SLA generation

The prime responsibility of SLA determination is to generate a SLA ID for a payment order that results in triggering different processing conditions for a payment. SLA Determination will be done only once for a payment in its life cycle. The determined SLA will be used mainly to determine:

The debit and credit bank conditions.

Bank charges for the payment.

Influence of other factors on SLA

Code words

The sending bank, wishing to use a specific level agreement for a particular message, will specify the agreed code words in the payment message. SLA Determination component will determine the SLA reference for the payment from the code word provided in the message. When there is no code word in the message, the generic SLA agreed with the counterparty bank if exist will be used or the default SLA will be used for the payment. SLA configurations are specific to a company.

Note!

The SLA determination is always based on the original code words supplied with the payment order message along with the message priority for a company. It is never based on the result of the code word determination.

A correspondent bank may supply multiple code words in a message, but only a single SLA can be applied to a single payment. The SLA configuration will allow to define priority of possible SLA's to be specified in such cases. Single or a collection of code words will lead to one SLA based on the priority of SLAs.

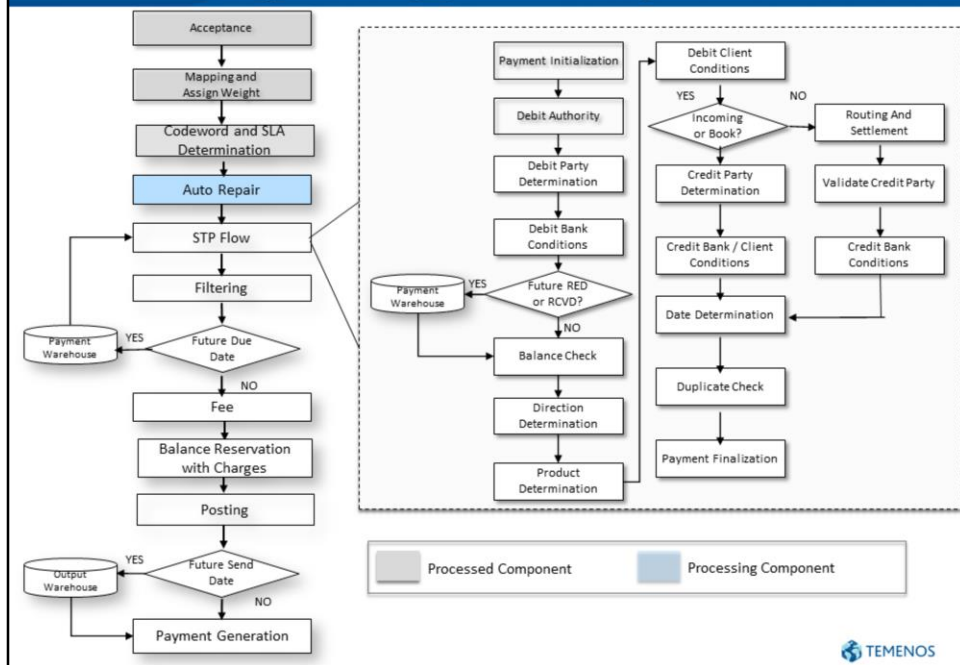
Weight

SLA determination may not be required for all type of messages. This can be controlled in this component by the weight assigned for the payment by the Assign Weight component. In this case SLA for the payment will not be populated.

For most payments the decision to continue SLA determination or not can be decided based on the Overall weight of the payment. However, in certain cases overall weight might not be the right call to make the decision. So the combinations of Overall weight and Specific Weight will be used to continue the SLA determination process

Model Bank > PAYMENTS > TPH > Parameters > SLA per CodeWord > SLA per CodeWord List

How does TPH process a payment message?



For a payment to be processed, number of internal activities need to happen on the payment. The activities are listed and each of them are impact payment processing in a specific way. This is explained in detail as we proceed.

Auto repair and Enrichment

- Increasing the STP process
- Integrated with External Automated Repair Engine
- Nature of enrichment
- Reduce the manual interventions
- Apply Fees depends on enrichment nature
- Applicable for STP payments



TEMENOS

TPH has now been Integrated with external Automated Repair Engine - STeP using Integration Framework

STP Payments in TPH can now be sent to Auto Repair Engine – STeP for Enrichments based on Rule Engine built inside STeP. This would facilitate automatic repair of STP Payments with Erroneous Inputs reducing manual Intervention and thereby Increasing the STP Hits

TPH has the ability to send Payment Instruction Details to an external Auto Repair Engine - STeP for enrichment and then take the enrichments back and continue processing the Payment Instruction.

When enrichment happens, it should be possible for the User to see the changes introduced in Audit Trails. The return codes indicating the nature of enrichment should be visible in an Audit Trail and they should be available for decisions on repair-fees

Integration of TPH with External Automated Repair Engine STeP will help reduce the % of manual Interventions for STP Payments

Erroneous Data in the Incoming Messages

- Take a look at the Tag 59 of Incoming MT101 Message having spaces in Beneficiary Account Line. This Transaction will Land in Repair due to Simple Reasons - TPH will fail to Identify the Account for Credit because in TPH we would have stored the IBAN without spaces which is the correct way of representing an Account / IBAN

```
{1:F01DEMOGBXAXXX2769820912}{2:01011235041221ABNANL20XXX03520236480411171235N}{3:{101:XXX}{108:ICM041222AIO}
:20:201506230001
:28D:00001/00001
:30:150617
:21:20150623000101
:32B:GBP99,00
:50H:/77488
A. Mike, Rotterdam
Edinburgh
United Kingdom
:57A:HSBCGB20
:59:/GB29 RBOS 6016 1331 9268 19
Ron Marcus
United Kingdom
:70:REMITTANCE INFORMATION
:71A:SHA
~}
```

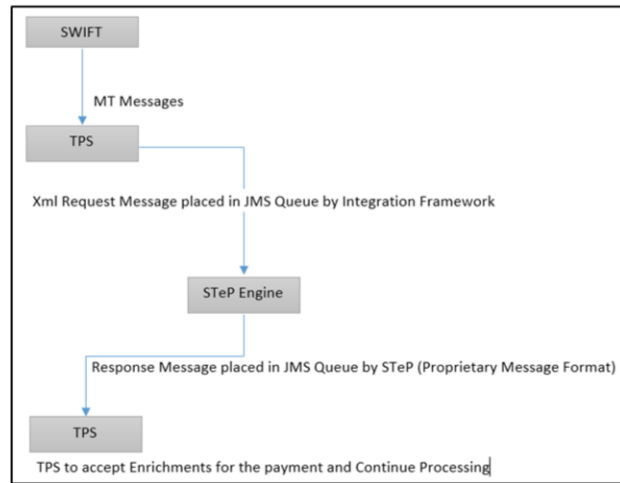
Erroneous Data in the Incoming Messages

- Data for Beneficiary in the SWIFT Message Received. Observe that Account Number is in Free Line 4
 - PARTYROLE = "BENFCY"
 - PARTYFREELINE1 = "JOHAN SMITS"
 - PARTYFREELINE2 = "PARKSTRAAT 5"
 - PARTYFREELINE3 = "AMSTERDAM"
 - PARTYFREELINE4 = "ACC NO 12312312312"
- Image after Enrichment by Automated Repair Tool. Account Number is copied to the appropriate placeholder
 - PARTYROLE = "BENFCY"
 - PARTYACCOUNTLINE = "/12312312312"
 - PARTYFREELINE1 = "JOHAN SMITS"
 - PARTYFREELINE2 = "PARKSTRAAT 5"
 - PARTYFREELINE3 = "AMSTERDAM"
 - PARTYFREELINE4 = "ACC NO 12312312312"

Erroneous Data in the Incoming Messages

- Data for ACWINS in the SWIFT Message Received. Observe that BIC is not Input
 - PARTYROLE = "ACWINS"
 - PARTYFREELINE1 = "RABO BANK"
 - PARTYFREELINE2 = "AMSTERDAM"
 - PARTYFREELINE3 = "NETHERLANDS"
- Image after Enrichment by Automated Repair Tool. BIC is auto-populated by Automated Repair Engine
 - PARTYROLE = "ACWINS"
 - PARTYIDENTIFIERCODE = "RABONL2AXXX"
 - PARTYFREELINE1 = "RABO BANK"
 - PARTYFREELINE2 = "AMSTERDAM"
 - PARTYFREELINE3 = "NETHERLANDS"

TPH-STeP Flow



Auto repair and Enrichment

Admin Menu > Payment Hub > Static Data GUI > Programs Per Weight

ProgramsPerWeightList

Component Name and Weight	Company Id	SW Code	Ranking	
AccountCashPosition.H	BNK	*	1	☒ ☐ ☐
AccountCashPosition.L	BNK	ICD	1	☒ ☐ ☐
		CDR	2	
		CCR	3	
		*	10	
AccountCashPosition.M	BNK	CQE	3	☒ ☐ ☐
		CQS	4	
		*	5	
Autoform.H	BNK	*	10	☒ ☐ ☐
Autoform.L	BNK	CDR	3	☒ ☐ ☐
		STL	9	
		ICD	10	
		CCR	11	
		*	20	
Autoform.M	BNK	CQE	29	☒ ☐ ☐
		CQS	30	
		*	31	
AutomatedRepairTool.H	BNK	*	10	☒ ☐ ☐
AutomatedRepairTool.L	BNK	CDR	4	☒ ☐ ☐
		STL	9	
		ICD	10	
		CCR	11	
		*	40	

First Level to enable Automated Repair Functionality in TPH is to have Programs Per Weight Configured for STP Payments. Depending on the requirement, User can enable it for Heavy Weight Payments Or Light Weight Payments. Programs Per Weight Records for both Heavy Weight and Light Weight is delivered for Automated Repair with Program Skip Indicator as 'N' which means Automated Repair Tool is enabled. The user has to Authorize these records to complete Programs Per Weight Setup. If Automated Repair Functionality is not required for STP Payments, then set Program Skip Indicator as 'Y'. (Please note that Programs Per Weight Records are Company Specific and this procedure has to be repeated for the Companies that require Automated Repair Functionality)

Programs Per Weight Table

ProgramsPerWeight AutomatedRepairTool.L

Amend Audit

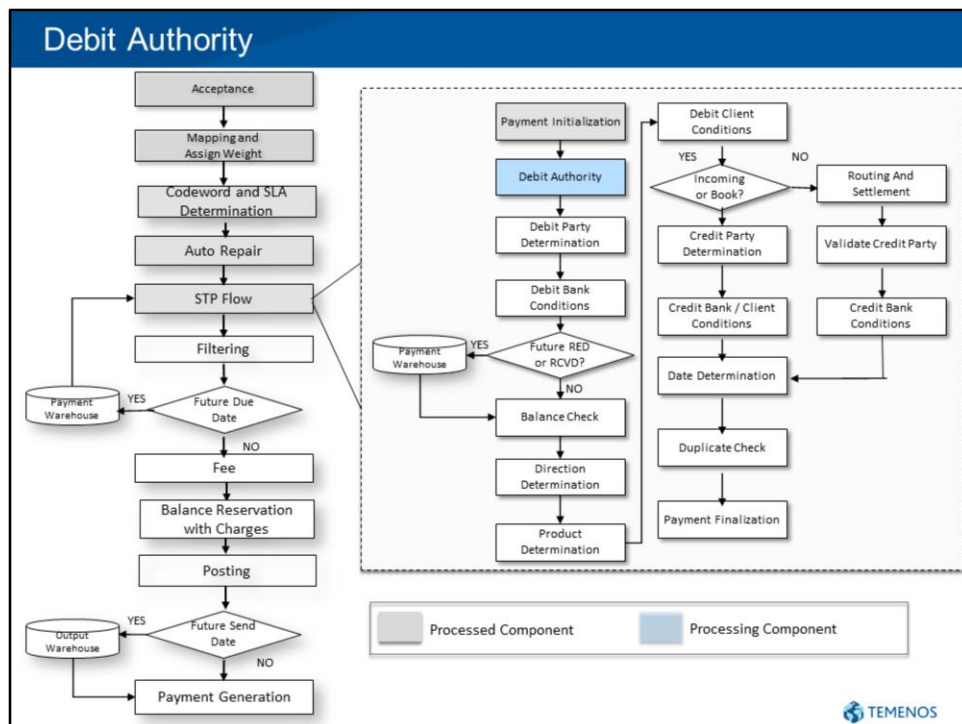
Company ID: SNK

Specific Weight Code 1: CDR
Ranking 1: 4.00
Program Name 1: AutomatedRepairToolService.repairPayment
ProgramsPerWeight Description 1: EN
Program SkipIndicator 1: ☒ Y ☐ N

Specific Weight Code 2: STL
Ranking 2: 9.00
Program Name 2: AutomatedRepairToolService.repairPayment
ProgramsPerWeight Description 2: EN Skip Program
Program SkipIndicator 2: ☒ Y ☐ N

Specific Weight Code 3: ICD
Ranking 3: 10.00
Program Name 3: AutomatedRepairToolService.repairPayment
ProgramsPerWeight Description 3: EN
Program SkipIndicator 3: ☒ Y ☐ N

Continuation screenshot of previous slide



Purpose

The Debit Authority component is responsible for verifying whether authority can be granted to payments for which the debit party is in the books of the processing bank. This is to prevent fraudulent behaviour from third parties as they have to specify the correct debit party/account in the payment message. By interpreting various payment characteristics the DA component is able to determine whether authority can be granted to the transaction.

Overview

Debit Authority is required for a Credit Transfer for which the debit party resides in the books of the processing bank requires a valid netting agreement. A netting agreement is a 3-party agreement between the sending bank, the processing bank and the customer. It states whether another bank is allowed to request a payment transfer, or send a payment instruction, specifying the debit account of the customer that is serviced by the processing bank. It is the responsibility of Debit Authority to validate whether the party requesting a payment transfer, or sending a payment instruction, has the authority to mention a specific third party as the debit party for the payment.

Direct debits will use DD.MANDATE to record and use mandates.

Debit Authority

Admin Menu > Payment Hub > Debit Authority GUI > Netting Agreement / No DA List

Payment Hub	^
Static Data GUI	▼
Authorization Principle	↻
Message Acceptance GUI	▼
Payment Router GUI	↻
Message Mapping GUI	▼
Batch Server GUI	↻
Local Clearing GUI	▼
Trip Status Action	↻
Payment Workflow GUI	▼
Weight Assignment	↻
Inbound CodeWords GUI	▼
Outbound CodeWords GUI	▼
SLA Per Codeword	↻
Bank Conditions GUI	↻
Debit Authority GUI	^
Netting Agreement	↻
No DA List	↻

ID	101BOFAUS33-20180316
Field Name	Field Value
Company	BNK
Message Type	101
Sending Bank	BOFAUS33
Debit Account	*
First Name/Address Line	*
LinkID	101BOFAUS33*-20180316
Start Date	16-Mar-18
End Date	31-Dec-99

Table	No DA List
ID	101-20160101
Field Name	Field Value
Company Code	BNK
Start Date	16-Mar-18
End Date	31-Dec-99

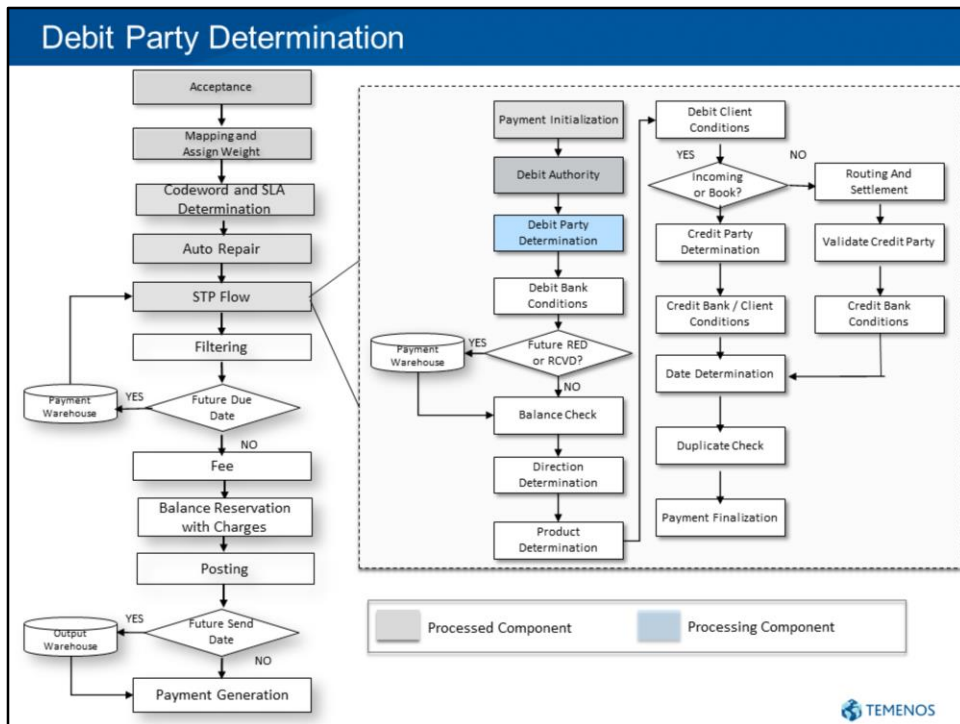
TEMENOS

Netting Agreement

Store the netting agreements which usually state which sender of the payment instruction has the authority to mention a third party as the debit party for the payment here. Whenever a message type of a company is not specified under NoDA list, a lookup is performed on this table for the type of message received from the specific sending bank. The agreement also holds the exact debit account in the processing bank for which a debit instruction can be carried out by the sending bank.

NoDA List

Define the list of message types of a company which do not require any authority to be processed here. If a record is present in this table, the payment will be authorized straight forward. If not, the transaction is not authorized directly. Rather the transaction is subjected to further processing such as a look up for valid agreements between the sending bank and the processing bank to authorize such payments.

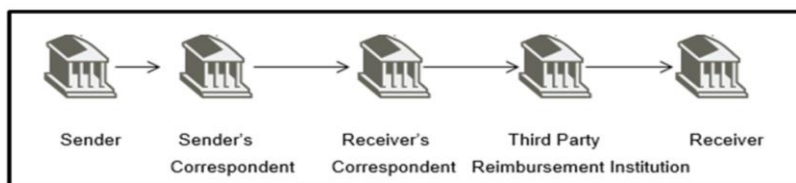


PURPOSE:

- Determine the correct debit account based on the information contained with the debit parties present in the payment instruction. If the account is already determined, the process is skipped.
- Pass debit account details to Account and Customer interface for validation.
- Interpret and update the payment file according to the output returned by the Account and Customer interface.
- Delete debit charge details only on request from STP flow.

Debit Party Determination ensures that the right debit account is assigned in the payment file. If the debit account is already determined, the determination process can be skipped and the account can directly be validated with the Account & Customer database.

Determining Debit Party



- Determine the debit party, determine and validate account
- Implied Debit – Validate account

As the name implies, the next step is to determine the party to be debited. For instance, when processing a 103, it is imperative to note that the ordering party has already been debited in the books of the ordering party's bank. When the 103 reaches us, we need to know whom we need to debit in our books. A 103 can be sent to us directly from the sender or it can be sent via various parties as listed below.

While determining the debit party, system will always check if the nearest party to the receiver is the actual party to be debited. Hence, will check

If a Third party reimbursement institution is present in the message in tag 55A, then, he is the party to be debited. Search for debit party ends here.

If Receiver's correspondent is present in the message in tag 54A, then, he is the party to be debited. Search for debit party ends here.

If Sender's correspondent is present in the message in tag 53A, then, he is the party to be debited. Search for debit party ends here.

If the account of the sender is present in the message in tag 53B, then, that is the account to be debited. Search for debit party ends here.

If none of the above is present, then, sender of the payment (Which will be available in the header) is the party to be debited.

Whenever a BIC is determined as the debit party (In cases 1,2, 3 and 5), it is assumed that the determined debit party has either a LORO or a NOSTRO relationship with us.

Hence, we extract the appropriate account for the BIC and mark that as that as the debit account.

In our case, as part of the message, we neither have 55A, nor 54A, nor 53A nor 53B. Hence, system would extract the sender from the message header and obtain the LORO/NOSTRO account of the sender

To determine the debit party, no specific configuration is required expect the definition of the appropriate LORO or NOSTRO accounts (using table PPL.LORONOSTROACCOUNT)

It is also possible for payments to come in with a forced debit party/account. This is done during the mapping phase of the payment. In such a case, validation of the debit account is what is required rather than determining the debit account.

Validating a debit account :

Account exists in PPL.LORONOSTROACCOUNT

Account exists in the DDA

Account is active and not closed

Account does not have any debit posting restrictions

Defining the LORO/NOSTRO Accounts

Admin Menu > Payment Hub > Routing and Settlement GUI > Loro Nostro Account

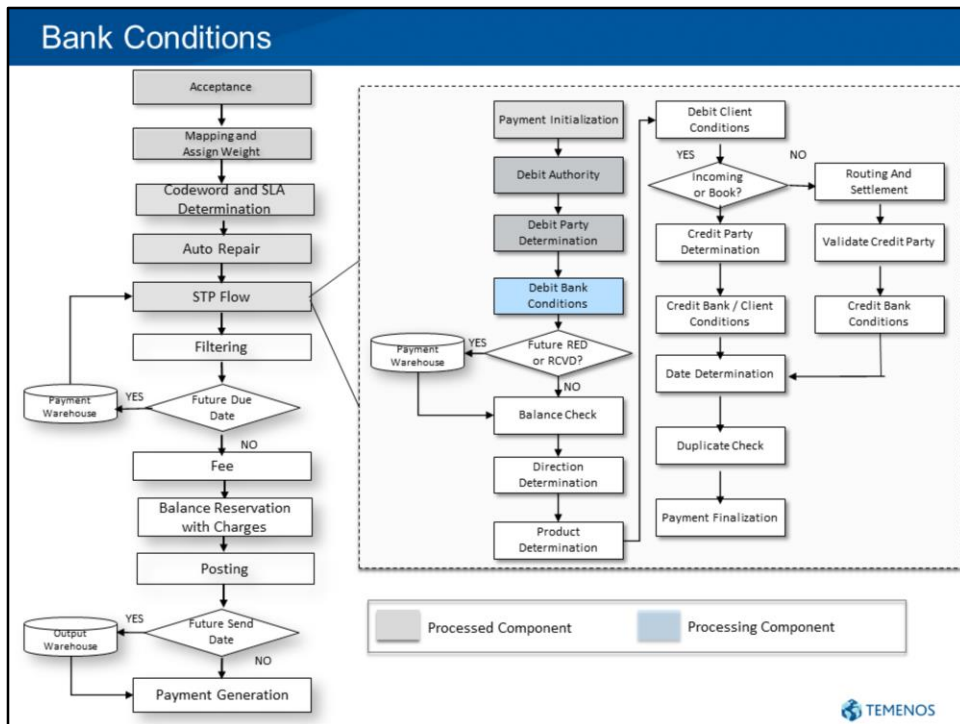
Loro Nostro Account 21-20180316

Company ID	BNK		
Party BIC	HSBCSGG		
Type of Account Number	N	Account Currency	SGD
Start Date	16 MAR 2018	End Date	31 DEC 2099
LinkID	HSBCSGSGGDN-20180316		

Account Details Audit

AccountNumber/CompanyID	AccountNumber*	OwningBIC*	PreferredDebitAccountNumber*	PreferredCreditAccountNumber*	ChargeIndicator*
BNK	20796	HSBCSGG	☒ Y ☐ N	☒ Y ☐ N	☐ Y ☒ N

Store the accounts of the correspondent banks with which the bank shares a LORO and NOSTRO relationship here. This table is linked with the BIC table. As the BIC table is company specific, this table is also company specific. This table is also used in the credit account determination step.



PURPOSE:

Bank Conditions or Correspondent bank conditions, define the way in which a payment should be processed in the payment engine for a correspondent bank. Bank conditions of the correspondent bank involved in the payment will influence mainly **Straight through processing** of the payment in the payment engine. This will also allow customizing banks' charge account, statements and advices. The Bank conditions component allows the behaviour of the payment engine to configure at a flexible level where debit or credit party (or both parties) in the payment is a bank.

In simple terms, the scope of the Bank conditions is to define the conditions for warehouse, advices, charge account, statement narrative in postings and STP flow.

OVERVIEW:

- Special conditions can be agreed and applied on the payments for specific sending or receiving banks when payments are received from or sent to them respectively.
- The bank can have a generic conditions and/or Service Level Agreement (SLA) based conditions with the correspondent banks.
- Bank conditions can be applied at a number of different levels, ranging from bank (BIC) code to combinations of bank, specific SLA's and transaction currency.
- If no agreement exists between the banks then default bank conditions will be applied on the payment.
- In case of customer transfer, bank condition will be applicable for the bank side of the payment where bank is involved.

Bank Conditions

Admin Menu > Payment Hub > Bank Conditions GUI

Payment Hub

Static Data GUI

Authorization Principle

Message Acceptance GUI

Payment Router GUI

Message Mapping GUI

Batch Server GUI

Local Clearing GUI

Trip Status Action

Payment Workflow GUI

Weight Assignment

Inbound CodeWords GUI

Outbound CodeWords GUI

SLA Per Codeword

Bank Conditions GUI

Bank Conditions Amend DEFAULT-20160101

Bank Conditions
BankCondChargeAccount
BankConditionsAdvice
Audit

Company

Correspondent BIC/HIC

SLA ID

Currency

Start Date

End Date

Non STP for CTR

Credit Special Instruction

Non STP for BTR

Debit Special Instruction

Warehouse Indicator

Language

Debit Statement Format

Credit Statement Format

FX Spread

LINKID

AllowSpecialCharacterSet

CodePageSet

Tran Ack Hack Indicator

Interim Status Indicator

Customer Status Msg Type

☐ Y ☒ N

☐ Y ☒ N

☐ Y ☐ N ☒ J

☐ Y ☐ N

☐ Ack ☐ Hack ☐ Both

☐ Y ☐ N

CompanyID : It is the company code for which the Bank Conditions record is created.Valid value from PP.COMPANY table .

CorrespondentBIC : This field refers to the BIC corresponding bank (Sender or Receiver).

Currency Code : It refers to the transaction currency of the payment. Valid value from PP. Currency table OR '*' ('*' value means it can match all the currency code).

Start Date Bank Conditions : It is the date from when this record is active.

Credit Special Instruction : It defines the text that will be displayed on the repair screen when the payment is in repair queue. This text in this field will be displayed when bank conditions is read for the receiving bank (credit side).

Debit Special Instruction : It defines the text that will be displayed on the repair screen when the payment is in repair queue. This text in this field will be displayed when bank conditions is read for the sending bank (Debit side).This field can only be filled when BTRNonSTPIndicator is set to 'N'

LanguageID : This indicates the language the corresponding bank would like to have the statement narratives (including charge description) in SWIFT advice and account statements. If this field is empty, then the language defined at company level will be used. Valid value from PP.LANGUAGE table or blank.

CreditStmntFormatName : This field indicates the preferred statement format to use when the corresponding bank's account is credited. Statement format defines a set of statement narratives that will be used when an account is debited/credited in the ledger. If this field is empty, then the statement format defined at posting line level or company level (if not defined in posting scheme) will be used. Valid value from PP.STATEMENT.FORMAT OR blank.

DebitStmntFormatName : This field indicates the preferred statement format to use when the corresponding bank's account is debited. Statement format defines a set of statement narratives that will be used when an account is debited/credited in the ledger.If this field is empty, then the statement format defined at posting scheme or company level (if not defined in posting scheme) will be used. Valid value from PP.STATEMENT.FORMAT OR blank.

FXSpread : This field defines the preferential rate to be used while determining the spread for a currency exchange. The rate mentioned is a percentage of the standard spread.

EndDateBankConditions : It is the date till when this record is active.

ChargeAccountIndicator : This field indicates that Bank requires separate charge account or charge accounts for different transaction currency. This field is only required for GUI validation. It is not stored in database. Y or N

TransactionCurrency : This field indicates the currency that will be the transaction currency for the payment. Valid entry in PP. Currency table or '*'

ChargeAccountCompanyID : This field indicates the company ID of the Charge Account Number defined.

This should be a valid record in the PP.COMPANY. Input to this field can only be provided if Separate charge Account Indicator is set to Yes

ChargeAccountNumber : This field indicates the account number from which the charges has to be collected.

ChargeAccountCurrency : This field indicates the currency of the Charge Account Number defined. This should be a valid currency code. If left blank, this will be wild carded.

AdviceIndicator : This field provides details on the applicability of AdviceIndicator Y or N

SequenceNumber : This field is system generated

DebitCreditAdvice : This field indicates whether a debit or credit or Charge Advice confirmation needs to be generated.

D(debit)

C(credit)

CH(Charge Advice)

Input to this field can only be provided, if Advice Indicator is set to Yes

CTRBTRIndicator : This field defines the transfer type of the payment.

Transfer type can be customer or bank transfer.

C - Customer Transfer

B - Bank Transfer

InitiatedByOthers : This field indicates if Advices are generated by Bank Conditions or can be initiated by others.

Y (Yes)

N (No)

B (Both)

AmountCurrency : This field indicates the currency that is associated with the amount

FromAmount : This field along with ToAmount field indicates the amount range of the transaction for the product.

This field indicates the start amount of the range.

Number must be defined.

Default : 0

ToAmount : This field along with FromAmount field indicates the amount range of the transaction for the product.

This field indicates the end amount of the range.

Range is defined in the transaction currency

Number must be defined.

Default : 9999999999999999.00

DeliveryMethod : This field indicates through which channel advice can be sent to customer
SWIFT / POST / EMAIL / PHONE / FAX / SMS

Telephonenumber : This field holds the phone number, used for confirmations

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'Phone'

EmailID : This field holds Email address to send advices to customer through Email

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'EMAIL'

BICAddress : This field holds the valid BIC

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'SWIFT'

SMSNumber : This field holds Phone number to send advices to customer as SMS

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'SMS'

FaxNumber : This field holds Fax Machine Number to send advices to customer through FAX.

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'FAX'

PostName : This field holds the street address2 and advices generated will be sent to through Postal

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'POST'

PostAddress1 : This field holds the street address3 and advices generated will be sent to through Postal

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'POST'

PostAddress2 : This field holds the street address4 and advices generated will be sent to through Postal

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'POST'

PostAddress3 : This field holds the street address4 and advices generated will be sent to through Postal

Input to this field can only be provided, if Advice Indicator is 'Yes' and Delivery Method is 'POST'

Attention : Indicates useful information relating to the delivery methods and the person whom to contact.

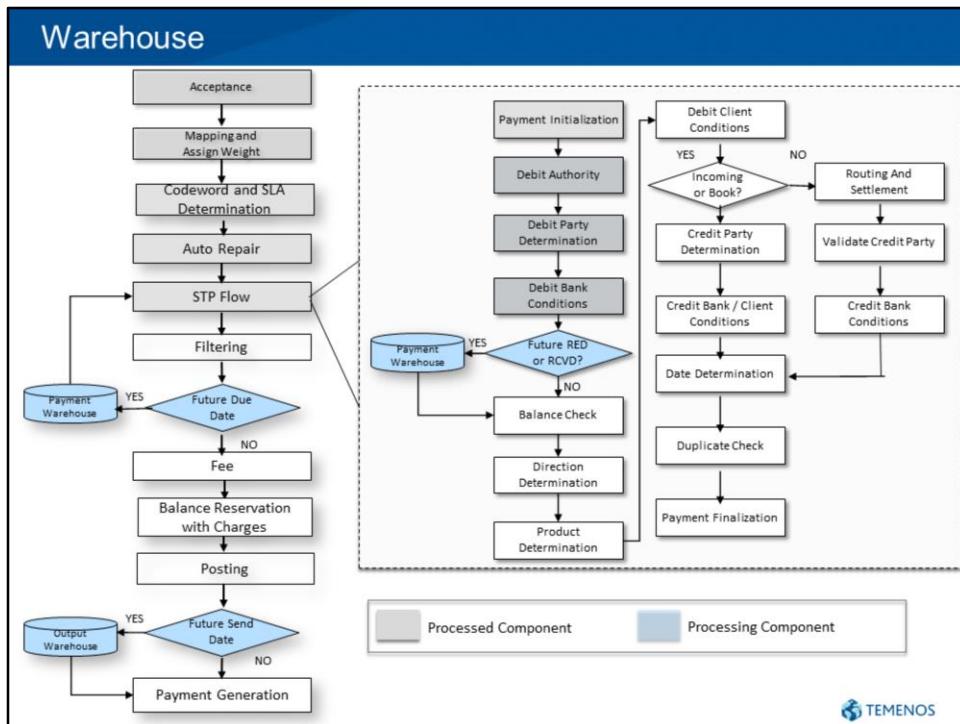
E.g. In case of a Phone confirmation, the attribute can hold information on the name of contact and the preferred contact time and/or the language.

AllowSpecialCharacterSet : This field indicates if special characters are allowed Y or N

CodePageSet : This field specifies against which code page the special characters have to be validated. The value inputted by the user in this field will be validated against the ASCII.VAL.TABLE

Possible Values:

STANDARD.SW for LATIN or STANDARD.GR for GREEK



Banks provide Customers / Corporates an opportunity to schedule future payments in advance. These payments are held by the bank and processed in time, honouring the future date as requested by the customer.

The customers get the dual benefit of scheduling a future payment in advance, along with the flexibility to cancel / make changes to these payments (until the actual due date).

There are some Clearing systems which do not accept future dated payments and some which only accept payments certain days in advance. (E.g. EBA does not handle future dated payments and TARGET2 can only handle payments up to 5 days in advance).

In such cases, even though the payment has undergone complete payment processing and is booked in the Ledger, payment message cannot be sent to the clearing system to avoid rejection.

In both the above scenarios, the payment engine has to undertake the responsibility of housing these payments. Warehouse functionality within the payment hub that holds these payments that need to be processed at a future date.

Types of Warehouse



- Payment warehouse – When Requested Execution Date or Requested Credit Value Date is in future, payments are warehoused at the start of STP flow based on Bank Conditions
- Future due date warehouse – When calculated Credit Value date is in future and processing date is also pushed to future to meet the credit value date
- Send date warehouse – When Send date is in future



How does Dates impact Payment Warehousing?

Payment Warehousing is performed when the payment has a future **Requested Due Date** or **Requested Credit Value Date**.

The difference between these two dates can be explained with the help of an example as below.

Requested Due Date

On 7th July, start of MediaMarkt wants to schedule the payment of its invoice received from Dell computers on 7th July 2012. On 1st July, a payment instruction is created which is scheduled to be executed on 7th of the Month (Thus 7th July = Due Date). The payment engine will send this payment to the warehouse. day, payment is released from warehouse for further payment processing.

Requested Credit Value Date

Albert Hein (AH) has ordered a shipment of 30,000 boxes of Kiwi fruit from a supplier in New Zealand. The shipment will happen only when the payment is made available in the supplier's account on 7th July (Thus 7th July = Credit Value Date).

On 1st July, a payment instruction is created. The bank has to ensure that payment reaches suppliers account on 7th July. Therefore, payment engine processes the payment until routing and settlement is chosen for this payment (as opposed to warehousing it initially)

When does Payment Warehousing executed for a Payment?

Payment Warehousing is usually called for payment which consists future due date. As explained earlier, Payments with a **Future Requested Due Date** will be warehoused immediately on understanding that customer has requested a future execution date. i.e. Customer requests for payments with future due date are always warehoused.

And, Payments with a **Future Requested Credit Value Date** will be processed to an extent, so that R&S is able to figure the route, which is then taken into consideration while defining the Due Date / Credit Value Date.

In both the cases, such Payments are parked with Payment Warehouse.

Also, these payments will undergo complete STP re-processing as and when they are released from the warehouse.

Why does a Payment released from Payment Warehouse needs to undergo STP re-processing?

The reason being that, during the time the payment was in warehouse, certain payment processing information could have undergone change for e.g. the static data (BIC table), client condition, R&S agreements etc.

Thus, it is important to process these payments again.

When does Output Warehousing executed for a Payment?

Payments which are already processed by the payment engine and the corresponding posting entries are booked with future send date are mainly warehoused as the particular output channel does not accept future dated payments.

For Example,

V&D is a customer for SNS Bank in Netherlands. V&D initiates a cross border payment to Germany. However, SNS Bank does not have cross border payment system. They forward all their payments to ABN AMRO Bank (AAB) for processing.

The agreement with SNS bank is that, all payments will be processed and booked as and when received by AAB (no warehousing)

AAB receives the payment request on 2nd July from SNS bank with a requested CVD of 10th July. AAB process the payment until routing and settlement.

R&S figures out that payment is to be sent via EBA. However, EBA does not accept payments with CVD later than today. In such scenario, payment message cannot be sent today to EBA to avoid rejection.

Hence, payment is processed and booked today (2nd July = Book Date) but the payment message will only be sent on the requested CVD.

These payments are directly released to the **Output Warehousing** component.

How Debit Bank Conditions are associated with Warehouse?

The decision to warehouse payments varies from bank to bank.

A bank may choose to use the warehouse functionality for all payments or only for certain payment types.

Bank Condition decides whether to warehouse payments or not

If Bank condition says **NO Warehousing**, then payment will never fall in warehouse

Release of Payment from Warehouse

On the exact Release Date (i.e. Processing date / Send date), payments are released back to the respective payment engine flows / components.

Release of Payment to STP Flow

When the Processing Date $\leq$ Current Business Date of the processing company, At SOD the payment is released from warehouse back to the STP flow.

Release of Payment to Payment Flow / Output Generation

When the Send Date = Current Business Date of the processing company, At SOD the payment is released from warehouse back to Payment flow.

Release from Warehouse

- Moves Payments from Warehouse to STP, Filter and Payment Generation based on different Payment status
- Triggered during Start of Day (SOD)
- Release from warehouse – Normal and force release
- A warehoused payment can be cancelled except when it has been booked (A payment in output warehouse cannot be cancelled)
- The below menu option is used to release payments from the warehouse. User Menu > Payment Hub > Payment Exceptions > Warehouse and Parked Payments > Warehouse Payments List> Warehouse Payments List



When a payment gets processed by the payment engine, after the debit party is identified, the payment date would be checked to see if it is future dated or not. If it is future dated and the sender bank has opted to warehouse a future dated payment, system will mark the payment as '**To be warehoused**'.

Payment Warehousing - Releasing payments from warehouse on the due date for further payment processing.

Sending warehoused payments to the **Filtering** Module to perform daily checks

Output Warehousing - Releasing payment to the Clearing systems in order to honour the Credit Value Date (CVD)

Cancellation / Amendment of warehoused

Payments released from the warehouse will undergo complete STP re-processing as and when they are released from the warehouse. The reason being that, during the time the payment was in warehouse, certain payment processing information could have undergone change for e.g. the static data (BIC table), client condition, R&S agreements etc. Thus, it is important to process these payments again.

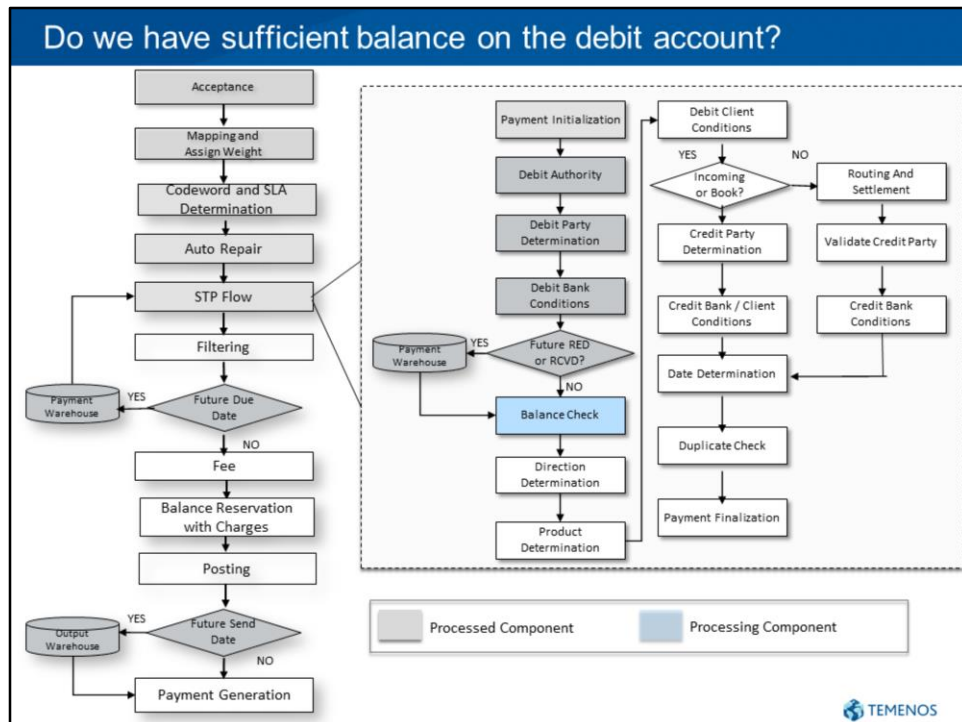


RED	RCVD	WH FLAG	SS	FXS	CS	PD	CVD	WAREHOUSE
14/11		Y	1	1	1			
15/11		Y	1	1	1			
	15/11	J	1	1	1			
	14/11	J	0	0	0			
	15/11	Y	0	0	0			
	15/11	J	1	0	0			
	16/11	J	1	0	0			
	15/11	N	1	0	0			



Take an example of the above transfers, you need to calculate the Credit value date and identify whether the payment will be warehoused or Not. Note that current business date is 14/11.

- RED- Requested Execution Date
- RCVD- Requested Credit Value Date
- WH Flag – Warehouse Flag Indicator
- SS – Settlement Shift
- FXS – FX Shift
- CS – Cutt off Shift
- PD – Processing Date
- CVD – Credit value Date



Purpose

Balance Check, as the name suggests, is considered as a vital component as it is imperative for the Bank to ensure that the Debit account has enough funds for the payment to be processed.

If the required balance is available for the transaction then the DDA (Demand Deposit Account) carries out a Reservation on the Debit account.

This Reservation implies that the Transaction amount is earmarked for the payment.

Balance Check

The Balance Check Component within the payment engine covers the functionality for two major facets

Funds Authorization

Funds Authorization is the process of checking whether the Debit Account has enough funds for the payment to be processed and if present then Reserving (Earmarking) the funds for the same.

Cancel Reservation

Cancel Reservation is the process of removing a reservation that is already applied on an account. This is required when a payment is Cancelled and it has a valid reservation applied on the debit account. Or in cases wherein the debit account is changed and the old debit account had a reservation applied on it.

Note!

The check for Debit Authority and Debit Party determination is completed before the call to the Balance Check Component. Balance Check Component will use the debit account determined by Debit Party Determination for the Reservation.

Balance Check

- Check for balance on debit account and debit charge account
- When reserving balance, a record in AC.LOCKED.EVENTS is created
- During posting, the reserved amount is utilised and hence the lock is removed.
- Pre-authorisation is possible. This would block funds on an account.
 - Pre-auth key this created, to be linked to the payment
- Credit Transfers
 - When a credit transfer is to be sent out, check if the debit account has sufficient balance to cover transaction amount or transaction amount plus charges
- Direct Debits
 - When a direct debit is received, check if the debit account has sufficient balance to cover the transaction amount or transaction amount plus charges



Funds Authorization – Funds Authorization is the process of checking whether the Debit Account has enough funds for the payment to be processed and if present then Reserving (Earmarking) the funds for the same.

The Balances and Limits (if applicable) for an Account are stored in the DDA (T24) and therefore a check on whether the Debit account has enough funds for the payment to be processed is carried out by the DDA. If the required balance is available for the transaction then the DDA carries out a Reservation on the Debit account. This Reservation implies that the Transaction amount is Earmarked for the payment. Any payment thereafter processed for this account would consider the earmarked amount while computing the available balances and limits.

The DDA also provides a provision with which a Manual Reservation can be carried out. This can be carried out in two scenarios –

When Funds Authorization request that was sent by the payment engine was rejected by the DDA and the CAO officer carries out a Manual Reservation for the payment.

CAO officer can choose to carry out a Manual Reservation even before the Funds Authorization request was sent to the payment engine.

The DDA provides a Manual Reservation Key for such reservations. This Manual

Reservation key as provided by the DDA can be entered on a payment through an OE / Repair screen of the payment engine. (This manual reservation key is called as Pre-Authorization key within the payment engine.) If provided through the OE / Repair screen then the DDA is only expected to validate this pre-authorization key with the Manual Reservation stored within its database. The check on limits / balances for the account were carried out at the time of making a manual reservation.

Cancel Reservation – Cancel Reservation is the process of removing a reservation that is already applied on an account. This is required when a payment is Cancelled and it has a valid reservation applied on the debit account. Or in cases wherein the debit account is changed and the old debit account had a reservation applied on it.

The payment engine will generate and send a Cancel Reservation request to the DDA. The DDA carries out the removal of the reservation and thereafter the payment engine should process the response from the DDA for that request.

Balance Check

AC.LOCKED.EVENTS

Repair View Screen | OE18107NNWMS

Payment Processing Information		Processing Company	BAK
Status	687	Processing Date	17 APR 2018 17 APR 2018
Transaction Reference Number	BNK18107KJCCDLF		
Sender's Reference Number	T24034490		
Priority	1		
Source	SWIFT		
Output Channel	LOBO		
Direction	D		
Transaction Currency	USD		
Transfer Type	C	Transaction Amount	3,500.00
Incoming Message Type	101	Charge Option	Sha
Balance	A		
Balance Reservation Number	ACLK1810766478		
		Sender Institution	BOFAPS33
		Receiver Institution	CRESCHZZ

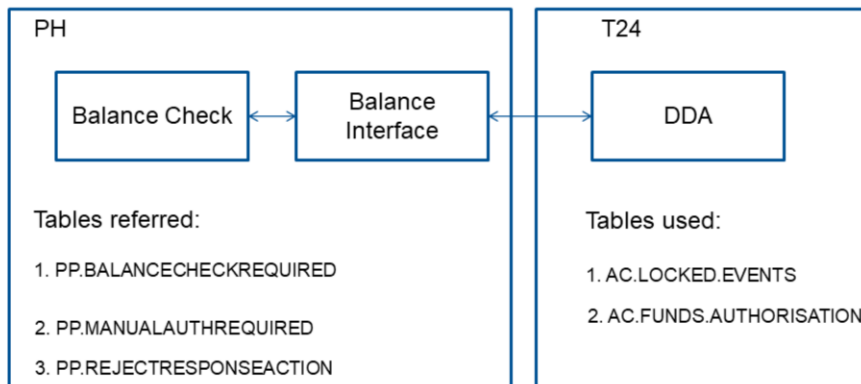
AC.LOCKED.EVENTS | ACLK1810766478

Account Number	11193	NRZ
From Date	17 APR 2018	17 APR 2018
Locked Amount	3,506.18	
Transaction No	BNK18107KJCCDLF	
Payment Amount	3,506.18	
Payment Currency	USD	US Dollar
Record Status	REVI	REVERSE Reversed
Quit No	1	
Inputter 1	47_RNFUTTER	
Date time 1	21_APR 18 18:50	
Authoriser	47_RNFUTTER	
Company	CB00010001	Master Bank



Field highlighted in the above TPH payment created the Locked amount in AC.LOCKED.EVENTS. The Record Id for AC.LOCKED.EVENTS is Balance Reservation Number of the payment.

Balance Check- Configuration Tables – Overview



TPH, when processing payments, has the capability to reserve the transaction amount on the debit account and then execute the various rules based payments processing. This capability to reserve the balance in an account can be controlled based on configuration and can be switched off too. For instance, it would not be required to perform balance check and reserve balance on a Nostro account if it is the debit account.

Balance Reservation is currently done for the Transaction Amount converted to the Debit Account Currency using mid-rate. If the debit account and the transaction amount are in different currencies, a mid rate is applied and balance is reserved. Balance is reserved by sending a request to the DDA. DDA then validates the account (checks for restrictions), checks for balance availability and if available, reserves the balances and returns the reservation key. TPH then processes the payment and after posting the accounting entries for the account, unwinds the reservation keys.

Balance Check- Configuration – Balance Check Required

- Check and reserve balance on debit account and debit charge account
- Admin Menu -> Balance Check GUI

Balance Check Required
PPBCR17621163495872102

Amend
Audit

Company
Bank -
Source -
Account Type -
Message Type -
Clearing Nature Code -
Settlement Transaction Indicator -

BNK
16
*
C
*
*
● * ○ Y

Balance Check Required Flag -
Reserve With Charges
OE Repair Reservation
Hold Bal For Future Processing Of
Approval Code: 1
Action: 1

● Y ○ N
● Y ○ N
○ Submit ● Authorise ○ STP
○ Y ○ N
- Please Select
○ Retain ○ Cancel

Turn on or turn off balance check using the table PP.BALANCECHECKREQUIRED. Do not turn on or off using PP.PROGRAMSPERWEIGHT.

Fields highlighted in blue influence balance check. Based on values in these fields, balance check can be turned on/off etc.

When processing local clearing payments, based on the Clearing Nature Code variations in balance check could be required.

When processing a settlement transaction, variation could be required in performing balance check. Some bank may not wish to perform balance check when processing settlement transactions.

All fields that influence balance check are available in POR.TRANSACTION.

Balance Check Required Flag – Use this field to turn on or turn off balance check.

Reserve with charges – If balance check needs to be performed, should be performed for the transaction amount or transaction amount plus charges.

OE Repair reservation – When an Order Entry (OE) payment or payment is repaired, this field specifies when balance check needs to happen on such payments. It could happen on submit, on authorise or during the STP flow.

Some banks would wish to send a payment to the authorisation queue only when sufficient balance is available. In this case, set this field to 'Submit'.

Some banks would want to check for balance only during authorisation. In this case, set this field to 'Authorise'

Usually this field is set to 'STP' when the system is implemented in a stand alone mode. In a stand alone implementation, during submit or authorise, an asynchronous call to an external DDA will not be possible and hence this option is made available.

Note : For standalone, this has to be set to 'STP'

Hold Balance For Future Processing Date – Assume that balance has been reserved for a payment. System then computes that the payment needs to be processed on a future date (Future processing date). In this case, payment would be sent to the 'Future due date warehouse'. When parking in the future due date warehouse, whether the balance that has been reserved should be retained or should be unreserved is controlled using this field.

ApprovalCode and Action:

Approval code indicates when payments are parked in the insufficient funds queue with a specific business state, such requests can be approved by a Credit Risk Officer or balance can be reserved by Recycler.

If the payment received with approval code, than Action field indicates whether the funds to be retain or cancel.

Balance Check with/without charges

Without Charges	With Charges
Performed after debit bank Conditions	Performed after FX and before posting
If FX is involved, mid rate will be used to convert transaction amount from transaction currency to debit account currency	If FX is involved, actual FX rate will be derived and transaction amount will be converted to debit account currency. Reservation will be done only if funds are available
During FX calculation, once actual FX rate is derived, if account balance is insufficient to cover transaction amount, account will be overdrawn.	When there is a FX threshold breach, then, mid rate +/- premium/discount is applied and then, the rate is arrived at.
Post balance check, if actual FX rates are higher than the utilised FX rates and account is short of balance, account will be overdrawn.	Post balance check, if actual FX rates are higher than the utilised FX rates and account is short of balance, account will be overdrawn.

FX threshold is defined in PP.COMPANYPROPERTIES

FX premium or discount is defined in PP.COMPANYPROPERTIES.

Balance Check- Configuration – Client Conditions

Client Condition Amend SEPADD_SET-20170213

ClientCondition Client Cond Advice Client Cond FXDiscount **Client Cond Charge Account** Audit

Separate charge Account Indicator * ☐ Y ☒ N

Indicator For Charge Account.1 ☐ C ☐ D ☐ A

Transaction Currency.1

Charge Account Company ID.1

Charge Account Currency.1

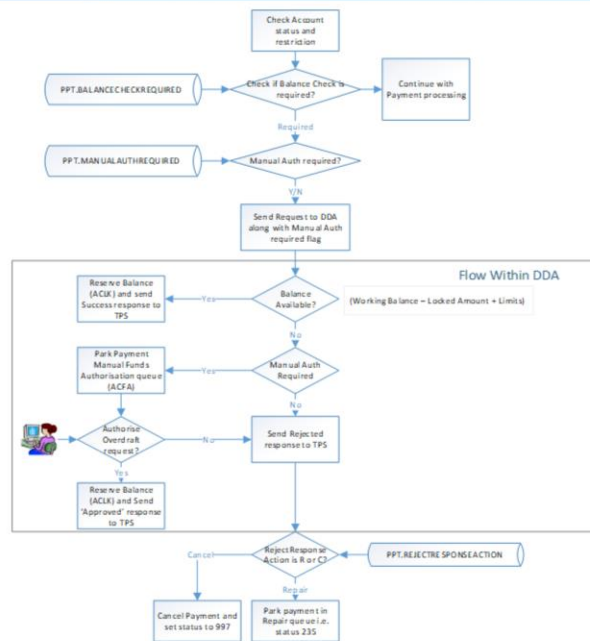
Charge Account Number.1

Balance Check On Chg Act ☐ Y ☐ N

If balance check has been configured to be done along with charges, then, this configuration allows us to define the following

1. If a separate charge account is used for charges, then, should balance check be done on the charge account
2. Note – If separate charge account is not configured and balance check is to be done with charges, then, this field will have no significance. Debit account will be checked to see if there is sufficient balance for transaction amount plus charges.

Balance Check Process Flow



The above diagram explains the process flow of Balance Check performed in Payments Hub.

Balance Check- Configuration - Manual Auth Required

Manual Auth Required
PPMAR17621163495872106
✓

Amend

Audit

Company	BNK
Rank *	40
Business Line *	*
Originating Work flow *	* ▼
Source *	* ▼
Message Priority *	* ▼
Banking Priority *	* ▼
Tran Amount Upper Limit *	* ▼
Message Type *	* ▼
Clearing Nature Code *	* ▼
Auth Required Flag *	☐ Y ☒ N

Manual Auth Required List

Define the Manual Auth Reqd Flag based on the characteristics of the payment such as Business Line, Originating Workflow, Originating Source, Message Priority, Banking Priority, Transaction Amount Limit, Incoming Message Type and Clearing Nature Code here.

The Manual Authorization Required flag is not checked for Funds Authorization requests with a Pre-Authorization key. Such requests are never sent for manual approval within the DDA.

If this flag is set to 'Y' then a Manual Authorization can be requested within the DDA if the debit account does not have the required balance for the reservation to go through.

If this flag is set to 'N' then a Manual Authorization cannot be requested even if the debit account does not have the required balance for the reservation to go through. Such a payment would get a 'Reject' response

The entries in this table are searched for a match in the order of its Ranking. The Manual Auth Reqd Flag is retrieved from the first entry that matches the payment characteristics. If no entry meets the match criteria then a default value of 'Y' is considered.

Balance Check- Configuration -Reject Response Action

Reject Response Action
PPRRA17621163495872200

Amend
Audit

Company	BNK
Rank *	40
Business Line *	*
Originating Work flow *	* ▼
Source *	* ▼
Message Priority *	* ▼
Banking Priority *	*
Tran Amount Upper Limit *	9999999999
Message Type *	* ▼
Clearing Nature Code *	* ▼
Manual Reject Response Action *	☒ R ☐ C

Reject Response Action

Defines the Reject Response Action Flag based on the characteristics of the payment such as Business Line, Originating Workflow, Originating Source, Message Priority, Banking Priority, Transaction Amount Limit, Incoming Message Type and Clearing Nature Code here.

The Reject Response Action flag is not checked for the responses received against a Funds Authorization requests with a Pre-Authorization key.

If this flag is set to 'R' then the payment for which a Funds Authorization response of 'Rejected' is received is to be sent to Repair.

If this flag is set to 'C' then the payment for which a Funds Authorization response of 'Rejected' is received is to be sent for Cancellation.

The entries in this table are searched for a match in the order of its Ranking. The Reject Response Action flag is retrieved from the first entry that matches the payment characteristics. If no entry meets the match criteria then a default value of 'R' is considered.

Note that the fields in this table are the same as the entries for the Manual Authorization Required table. Defining a new table for Reject Response action (and not using the same as Manual Authorization Required) gives the user an ease in operationally maintaining the tables

Balance Check – Insufficient Funds – Manual Auth Required

User Menu > Payment Hub > New Payment Orders > Payment Approvals > Manual Funds Authorisation > Request for Pending Approval

Unauthorised Manual Requests Results 1 - 1 of 1

Account Number: 77097 Currency: EUR
 Customer ID: 100160 Customer Name: Yhs
 Available amount: 10,000.00 Working balance: 10,000.00

Reservation Key	Payment Currency	Payment Amount	Reserved Amount	Excess amount	Value date	Created Date and Time	Decision	Decision Reason
BNK18107HDDOLMHRM	USD	15,000.00	13,971.79	-3,971.79	17 APR 18	17 APR 18 10:22:4446		- please select -

Unauthorised Manual Requests Results 1 - 1 of 1

Account Number: 77097 Currency: EUR
 Customer ID: 100160 Customer Name: Yhs
 Available amount: -3,971.79 Working balance: 10,000.00

Reservation Key	Payment Currency	Payment Amount	Reserved Amount	Excess amount	Value date	Created Date and Time	Decision	Decision Reason
BNK18107HDDOLMHRM	USD	15,000.00	13,971.79	-3,971.79	17 APR 18	10:22:4446	APPROVED	Funds expected to receive Overnight  

When a payments lands in the queue for manual approval, the officer can approve the overdraft or can choose to reject/cancel the payment.

Pre-Authorisation Key

Creation of preauthorisation key

User Menu > Payment Hub > New Payment Orders > Payment Approvals > Manual Funds Authorisation > Request for Pending Approval

Manual Requests Decision ACFA1810700815

Account Number	77007	Currency	EUR
Payment Currency	USD	Payment Amount	15,000.00
Value Date	2 JAN 17 A	Init Req Date	2 JAN 04 J
Init Req Time	15:08:14	Created Date	17 APR 2018
Created Time	10:22:44 AM	Transaction reference	99A101070000181M
Decision	Approved	Decision Reason 1	Funds requested to release Overdraft
Funds Auth Status	UNAUTHORISED	Reservation Key	ACFA1810700815
Request Source	PP	Reservation ID	ACFA18107003360
Request Code	01	Approval Code	☐ Manually authorised ☐ RECYCLER
Request Desc	No Funds		

Auth

Override 1	Account 77007 unauthorised overdraft of 1971.79 available 10000 Requested 13971.79 EUR
Override 2	Ac 77007 Today's Balance EUR 10000 will fall below Locked 13971.79 from 17 APR 2018
Record Status	INQJ
Current No	1
Inputter 1	99054,INPUTTER_OPS,ROSPA
Date / Time 1	04 APR 18 16:00
Authoriser	
Company	0000710001 Mexico Bank

Account 77007 unauthorised overdraft of 1971.79 available 10000 Requested 13971.79 EUR
Ac 77007 Today's Balance EUR 10000 will fall below Locked 13971.79 from 2 JAN 17 A

ID= RECORD WAS READ FROM HOLD FILE

CANCEL **ACCEPT**

Pre-authorisation key can only be used for OE and Repair payments as it needs the key to be keyed-in. It cannot be supplied as part of a message. When the account is insufficient funds the system throws override before you commit the record.

Using a Pre-Authorisation key

• OE screen with a Pre-auth key

The screenshot displays the 'Order Entry Book Transfer (CTR)' screen with the identifier 'OE18107N19G0'. The interface is divided into several sections for data entry:

- Payment Processing Information:**
 - Status: 135
 - Transaction Reference Number: [Empty]
 - Sender's Reference Number: [Empty]
 - Related Reference: [Empty]
 - Source: OE
 - Direction: B
 - Transfer Type: C
 - Incoming Message Type: RPCT
 - Balance Reservation: [Empty]
 - Balance Reservation Number: ACFA1810700815
 - Balance Reservation Detail Chq Acct: [Empty]
 - Pre-Authorization Number: ACFA1810700815** (This field is highlighted with a red box)
 - Intra Company Payment: ☐ Yes
 - Store Template Values: B-CRPCT-BNK-INPUTTER
 - Select from Template: [Dropdown]
 - Save As Template: ☐ Yes
 - File Name: [Empty]
- Processing Company:** BNK
- Processing Date:** 17 APR 2018 to 17 APR 2018
- Priority:** 1
- Product:** [Empty]
- Output Channel:** LEDGER
- Transaction Currency:** USD
- Transaction Amount:** 15,000.00
- Charge Option:** Ben, ☒ SW, Out
- Receiver Institution:** [Empty]
- Processing Date Imposed Flag:** ☐ Yes
- Output Channel Imposed Flag:** [Empty]
- Instructed Currency:** [Empty]
- Instructed Amount:** [Empty]
- Receiver Institution H/C:** [Empty]

When we try to process the payment with transaction amount greater than pre-authorization key amount than system will throw an error stating "Pre-Authorization Key is insufficient. Account could be overdrawn in case balance is insufficient."

AC.FUNDS.AUTHORISATION

AC.FUNDS.AUTHORISATION ACFA1810700815			Please Select	
Debit account	77097	YHSGG		
Account Ccy	EUR	Euro		
Requested Ccy	USD	US Dollar		
Requested Amt	15,000.00			
Request Date	2 JAN 17 J	2 JAN 17 J		
Dda Init Req Date	2 JAN 04 J	2 JAN 04 J		
Dda Init Req Time	15:58:14			
Orig Sys Date	17 APR 2018	17 APR 2018		
Orig Sys Time	101224446			
Orig Trans Ref	BNK18107000LMM			
Orig Trans Company	BNK			
Funds Decision	Approved			
Decision Reason.1	Funds expected to receive Overnight			
Funds Auth Status	AUTH000000			
Reservation Id	ACR1810703780			
Reservation Key	ACFA1810700815			
Request Source	PP			
Request Code	01			
Request Desc	No Funds			

Override.1	Account 77097 unauthorised overdraft of 3971.79 available 10000 Requested 13971.79 EUR	
Override.2	Ac 77097 Today's Balance EUR 10000 will fall below Locked 13971.79 from 17 JAN 2 JA	
Curr No	1	
Inputter.1	95954_INPUTTER_OFS_IRSPA	
Date time.1	04 JUL 18 16:09	
Authorizer	95954_AUTHORISER_OFS_IRSPA	
Company	GB0010001	Model Bank
Dept Code	1	Implementation

QUIZ

- Which component interprets the external payment messages from native format(i.e. SWIFT) to Neutral Payment Object?
- Which is the path through which the message reached TPH?
- What is called as specialisation of high level weight?
- Bilateral Codeword termed as “Banks / financial institutions can also make special agreements and come up with codewords to trigger bilateral agreements amongst themselves” – True or False

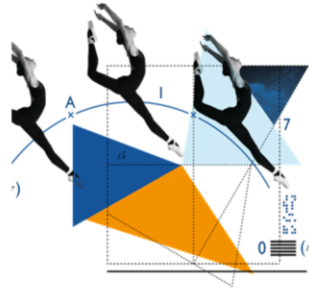
QUIZ

- Which component needs to be invoked for a payment leading to reduction in processing time?
- Name the table where the funds are locked for reservation of balance.
- Which component helps to store the future dated transactions and name the types.

What Did We Learn?

Conclusion

- Overview of payment processing in Payment Hub
- Static configurations used in Payment processing
- Process Incoming MT103 till Balance Check component



Thank You



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